Incremental Learning-Driven Segmentation and Clustering Optimization of UAV-LiDAR Rubber Tree Point Clouds

Hengrui Wang, Xiangjun Wang, Qin Zhang, Li Li, Jiang Zhang, Guoxiong Zhou, Yongfei Xue, and Haoyang Wang

***Abstract*—As an important economic crop in tropical regions, rubber trees require precise individual-level identification and structural parameter extraction to support productivity assessment and fine-scale plantation management. However, existing point cloud segmentation methods face significant challenges in handling species diversity, overlapping canopies, and clustering parameter sensitivity. Therefore, we propose an incremental learning-driven network, termed RtSegNet, for unmanned aerial vehicle-light detection and ranging (UAV-LiDAR)-based rubber tree point cloud segmentation and clustering optimization. First, we design a tree-structure incremental learning sampling (TSILS) strategy, which significantly improves model adaptability across diverse forest structures and enhances training efficiency. Second, a Fourier spectrum-enhanced spatial graph clustering (FSGC) method is proposed to improve feature discrimination in overlapping canopy regions. Finally, a Gaussian random duck swarm algorithm (GR-DSA) is proposed to adaptively optimize clustering parameters, enabling robust segmentation across trees of varying scales. Experimental results on a self-constructed rubber tree dataset show that RtSegNet outperforms ten state-of-the-art methods, achieving a mean intersection over union (mIoU) of 76.37%, an F-score of 87.11%, and reducing training time to 25.04 h. Generalization tests on the FOR-instance and NIBIO MLS public datasets further confirm the method's cross-environment robustness. Based on the segmentation outputs, key structural parameters such as tree height, canopy diameter, and canopy volume are extracted. The estimated tree height achieves an R^2 of 0.94 with a root-mean-square error (RMSE) of 0.52, providing strong technical support for accurate monitoring, yield prediction, health assessment, and the modernization of tropical agriculture. The source code in this work is available at https://github.com/aaaaasleep/RtSegNet**

***Index Terms*—Fourier spectrum-enhanced spatial graph clustering (FSGC), Gaussian random duck swarm algorithm (GR-DSA), incremental learning-driven, rubber tree point cloud segmentation, tree-structure incremental learning sampling (TSILS).**

I. Introduction

RUBBER tree is an important economic crop in tropical regions, with latex yield closely linked to tree health. Accurate identification of individual trees is essential for evaluating yield potential in rubber plantations. Precise monitoring at the single-tree level provides key morphological parameters [1] such as height, diameter at breast height, and crown width. These data support yield prediction models and guide precision planting layouts and health management strategies. Developing efficient segmentation technology for individual rubber trees [2] holds significant practical value and scientific importance for advancing modernization in tropical agriculture and ensuring stable development of natural rubber industry. Such technology facilitates automated yield monitoring systems through temporal crown analysis, enables early disease detection via morphological change assessment, supports precision fertilization strategies based on individual tree health metrics, and optimizes mechanized harvesting operations. These applications provide substantial economic benefits while promoting sustainable plantation management practices.

To overcome the limitations of 2-D tree imagery, researchers have turned to 3-D point cloud processing and proposed a variety of traditional algorithms [3], [4], [5]. Among these, watershed algorithms, prior knowledge-based methods, random forests, and graph-based segmentation have demonstrated varying performance across different tree species. Reitberger et al. [6] applied a watershed algorithm to generate canopy height models (CHMs) and identified stem positions via local maxima. They then used hierarchical clustering and the RANSAC algorithm to reconstruct tree trunks, followed by normalized cuts to segment small trees in 3-D. Heinzel et al. [7] developed a prior knowledge-based method

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for deciduous and mixed forests, where pre-classification of crown sizes served as prior information for individual tree extraction, significantly improving segmentation accuracy. Liu et al. [8] combined FiND techniques with random forests for boundary classification, enabling effective delineation of dense natural forest canopies with an overall accuracy of 82.4%. Strîmbu and Strîmbu [9] proposed a graph-based segmentation algorithm for extracting tree crowns from airborne LiDAR data. By employing a bottom-up strategy, this method quantified the topological relationships of canopy components in a weighted graph and ultimately achieved individual tree segmentation. Recently, Hui et al. [10] proposed an individual tree extraction approach for terrestrial LiDAR point clouds by integrating transfer learning and Gaussian mixture model (GMM) separation. This method combines deep feature transfer and probabilistic canopy separation, achieving robust performance in dense forest environments. Building upon this work, Hui et al. [11] further developed a multilevel self-adaptive individual tree detection method for coniferous forests using airborne LiDAR data. The multilevel adaptive approach effectively integrates raster-based and point-based segmentation, improving detection accuracy and adaptability under complex canopy structures.

Despite the progress of traditional algorithms in 3-D point cloud analysis, they often require meticulous parameter tuning and exhibit limited noise tolerance, making them less effective in complex forest environments. In contrast, deep learning approaches can automatically learn features from data, reducing reliance on manual parameter design while offering higher accuracy and automation in complex scenarios. PointNet [12] directly processes unordered point clouds using shared multilayer perceptrons (MLPs), capturing global and local features without requiring handcrafted feature projections, thereby improving computational efficiency. PointNet++ [13] extends this by incorporating hierarchical feature learning, recursively processing point sets in metric space to extract local geometric features at multiple scales. Li et al. [14] proposed the PointNLM network, which integrates point-wise and supervoxel-based representations to enhance segmentation performance. By modeling long-range relationships within point clouds, PointNLM achieved superior results in large-scale vegetation segmentation, reaching an IoU of 0.864 on the Semantic3D dataset. These results provide valuable insights for processing complex structures such as rubber trees. Building on these advances, more sophisticated deep learning models have been proposed to further improve segmentation accuracy and efficiency. Wielgosz et al. [15] introduced the Point2Tree framework for processing forest point cloud data. The framework first employs PointNet++ for semantic segmentation to classify point clouds into vegetation, stems, and debris, followed by a graph-based approach for individual tree instance segmentation. The method demonstrated strong adaptability and accuracy in forest environments with promising scalability. However, although the use of Bayesian strategies improved hyperparameter optimization, the method exhibited performance degradation in complex scenes. Kim et al. [16] proposed a new segmentation scheme for crowns, trunks, and branches by adapting the learning environment of the PointNet++ model. By introducing a resampling strategy, the method achieved an 11% improvement in accuracy and attained a mean $F1$-score of 0.86. Henrich et al. [17] proposed TreeLearn, a method that combines semantic and instance segmentation to effectively separate overlapping crowns and reduce dependence on handcrafted features. However, TreeLearn still faces challenges in offset prediction errors, particularly near crown boundaries or drooping branches, which may result in inaccurate point projections and lead to errors in initial clustering and instance assignment.

These studies have provided valuable insights into point cloud instance segmentation for rubber trees. Rubber trees exhibit distinct morphological characteristics that differentiate them from other forest species and create unique point cloud processing requirements. Unlike coniferous trees with symmetrical cone-shaped crowns and predictable branching patterns, rubber trees develop asymmetrical, umbrella-like canopies with irregular branching architectures. The systematic spiral bark tapping for latex extraction creates distinctive trunk scarring and influences branching development, resulting in uneven crown growth patterns not observed in natural forest species. In commercial rubber plantations, high-density planting arrangements create extensive crown overlap during mature growth phases, with adjacent tree canopies often merging into continuous vegetation masses. These species-specific characteristics necessitate specialized segmentation approaches capable of handling irregular boundary definition, variable point density distributions, and complex spatial relationships between adjacent trees. However, instance segmentation of rubber tree point clouds still faces three major challenges.

1) Tree species diversity is a key challenge in the segmentation of rubber tree point clouds. Distinct rubber tree varieties vary noticeably in geometric shape, branching patterns, and canopy structures, resulting in substantial interspecies variation in their point cloud distributions. This species-level diversity makes it challenging to establish a unified feature representation model for point clouds, as different tree types require different feature descriptors (as shown in Fig. 1).
2) The overlap of crown leaves is a significant interfering factor in the segmentation of rubber tree point clouds, originating from low shrubs and weeds (as shown in Fig. 1). In the intertwined canopy regions of neighboring trees, weakened gradient information and blurred canopy edge features make it difficult for segmentation algorithms to accurately distinguish between individual tree instances. This often leads to mis-segmentation or omission of trees.
3) The density of rubber tree point cloud data varies significantly, making it difficult to determine appropriate clustering parameters. Fixed parameters often perform inconsistently across regions with different point densities. Moreover, manual parameter tuning is both labor-intensive and prone to instability in clustering results, which ultimately reduces segmentation accuracy.

To address the challenge of tree species diversity, Liu et al. [18] proposed a single-tree point cloud segmentation technique

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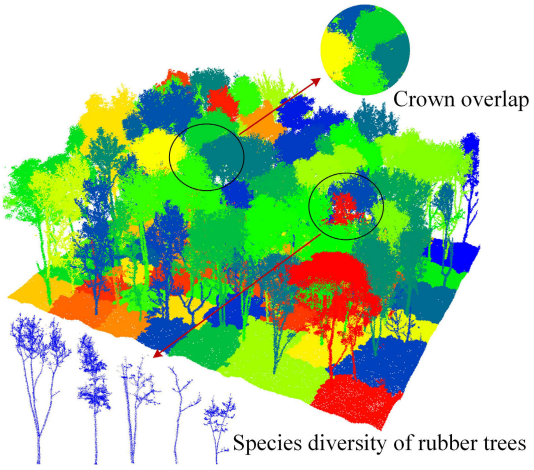


Fig. 1. Two challenges encountered in individual tree segmentation using unsupervised learning methods: morphological diversity of rubber trees and overlapping adjacent canopies.

based on a trunk growth (TG) approach. This method performs branch growth analysis using trunk points and assigns branch and leaf points through a nearest neighbor strategy to complete tree segmentation. While the method performs well across trees of different shapes and sizes, it incurs high computational complexity in regions with significant canopy variation and may yield segmentation errors in densely overlapping canopy areas. To further improve segmentation performance, Liu et al. [19] combined marker-controlled watershed segmentation with a spectral clustering refinement method. Tree apexes are identified by filtering local maxima, enabling coarse segmentation, which is subsequently refined using spectral clustering to reduce mis-segmentation in overlapping regions. However, the method still suffers from high computational costs, which may affect processing speed when handling large-scale data, particularly in complex canopy structures or dense forests. Zhang et al. [20] proposed an adaptive segmentation method based on extreme offset deep learning. The approach performs spatial transformations to shift each point closer to its tree apex, thereby enhancing intertree separability. An adaptive mean shift algorithm is then applied for further fine-grained segmentation. Although this method improves segmentation accuracy and shows strong adaptability in complex forest structures, it still faces challenges in balancing computational efficiency and accuracy when processing high-density forests or trees with large morphological differences. These limitations highlight the need for a more robust method capable of handling the segmentation of rubber trees with diverse structures.

To address the challenge of overlapping adjacent canopies, Li et al. [14] proposed a leaf segmentation method based on a 3-D joint filtering operator and patch-based region growing. The segmentation process was optimized using radius-based outlier filtering (RBOF) and surface boundary filtering (SBF), which effectively separated overlapping leaves and reduced the influence of noise. However, this method requires high-quality point cloud data, and its performance may degrade when processing low-quality or noisy inputs. Ma et al. [21] combined a local maxima algorithm with an improved watershed method to analyze the spatial density distribution of point clouds and effectively extract canopy regions. Although the experimental results demonstrated promising segmentation performance, variations in point cloud density led to suboptimal results in regions with complex canopy morphologies. Dai and Li [22] proposed a soft segmentation algorithm capable of automatically segmenting canopies and reconstructing canopy surfaces from LiDAR data. While the method performs well on large-scale forest point cloud datasets, it incurs high computational complexity and may result in long processing times. Therefore, it is necessary to propose a solution that balances efficiency and accuracy in handling canopy overlap.

To address the issue of oversegmentation and undersegmentation caused by the sensitivity to clustering parameters, a variety of studies have proposed different solutions. Ma et al. [23] compared several individual tree segmentation methods based on unmanned aerial vehicle-light detection and ranging (UAV-LiDAR) data, with particular attention to the impact of clustering parameters—such as minimum height, window size, distance threshold, and layer thickness—on segmentation accuracy. The sensitivity of parameter settings remains a critical challenge, especially in densely populated or high-density forests, where undersegmentation tends to occur. Li et al. [24] introduced a segmentation method that integrates a watershed algorithm with an optimized connected core evolution (CCE) clustering approach. Through an evolutionary process, clustering parameters are dynamically optimized, leading to improved segmentation accuracy. However, this method still suffers from reduced performance when applied to noisy or low-quality point cloud data. To further reduce this sensitivity, Zhang et al. [25] proposed a hierarchical filtering and clustering (HFC) algorithm for individual tree segmentation. This method enhances segmentation accuracy by applying multiscale clustering strategies and selecting adaptive parameters during the clustering process. While the HFC algorithm effectively mitigates the sensitivity to parameter settings, it exhibits high computational complexity when processing large-scale point cloud data. This is particularly evident in high-density forest environments, where runtime and resource consumption become significant. Therefore, it is necessary to propose a solution that further optimizes the automatic selection of clustering parameters.

Accordingly, the main contributions of this article are as follows.

1) To enable accurate segmentation of individual rubber trees and the extraction of their structural parameters, we constructed a dataset containing both rubber plantation and ground regions, in which each tree is assigned a unique instance label.
2) To address the issue of tree species diversity, we propose a tree-structure incremental learning sampling (TSILS) strategy, which adopts an incremental learning framework to retain previously learned knowledge while acquiring new tree species features, thereby reducing training time. Within this framework, a hierarchical tree-structure sampling method is designed to construct a multilevel feature exemplar library, further enhancing

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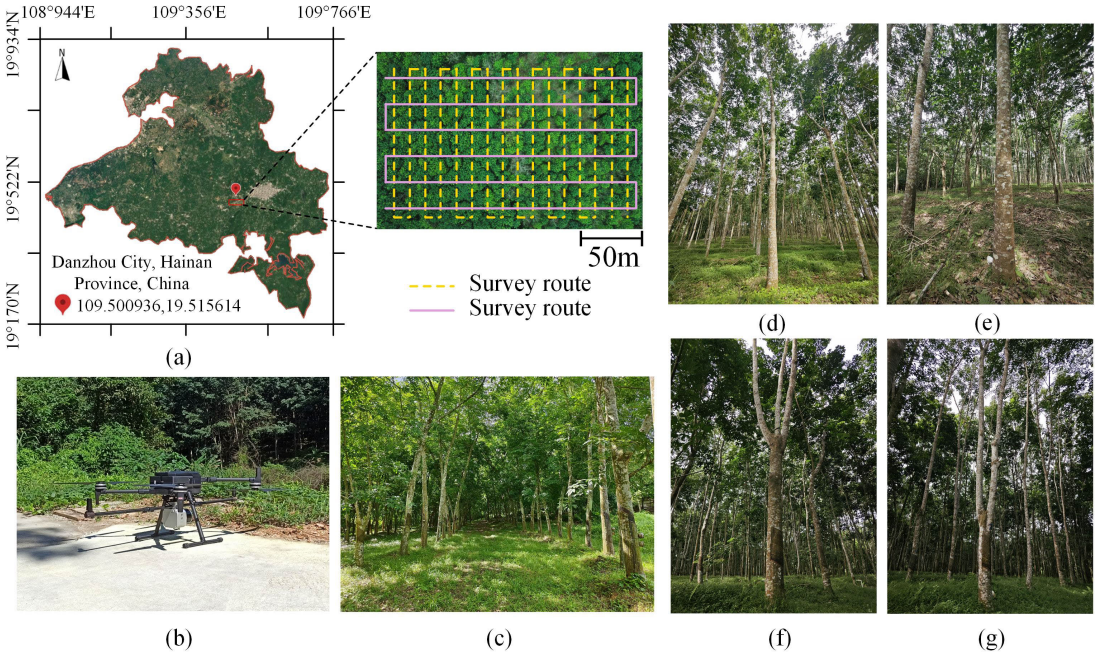


Fig. 2. Overview of the study area. (a) Location of the study site and UAV flight paths. (b) Data acquisition equipment. (c) Field photograph of the rubber plantation. (d)–(g) Detailed views of tree species within the plantation.

the model's generalization ability across diverse tree species.

3) To tackle the problem of overlapping canopy leaves, we propose a Fourier spectrum-enhanced spatial graph clustering (FSGC) method. After obtaining initial clustering results through region growing, a spatial relationship graph of seed points is constructed, and feature propagation is performed using a seed graph neural network (SeedGNN) to enhance boundary discrimination between adjacent trees. In addition, a Fourier spectrum merging strategy is introduced to further address the issue of overlapping branches and leaves between neighboring trees, effectively capturing the overall spatial topology of tree structures and improving segmentation accuracy in canopy-overlapping regions.
4) To determine the optimal clustering parameters, we propose a dynamic clustering optimization algorithm for rubber tree segmentation, termed the Gaussian random duck swarm algorithm (GR-DSA). This algorithm integrates swarm intelligence search with a Gaussian random walk (GRW) mechanism and constructs a cooperative search strategy with memory-based feedback. By dynamically balancing global exploration and local convergence based on individual tree sizes, GR-DSA automatically adjusts clustering bandwidth and sensitive parameters. This ensures accurate segmentation for both large and small canopies while maintaining adaptability across various rubber plantation scenarios.
5) To validate the effectiveness of the proposed method, extensive experiments were conducted. First, ablation and effectiveness studies were performed on RtSegNet to investigate the contributions of each module to the overall performance. Second, the proposed method was compared with several state-of-the-art point cloud segmentation networks and classical traditional algorithms to demonstrate its superiority. In addition, generalization experiments were carried out on the FOR-instance and NIBIO MLS public datasets to evaluate cross-scenario robustness. Based on the segmentation results, key structural parameters of rubber trees—such as height, canopy diameter, and canopy volume—were measured, with tree height estimation achieving an R^2 value of 0.94. This study introduces an innovative and robust framework for precise tree structure acquisition, contributing to advances in precision forest management and ecological condition monitoring.

II. Materials

A. Study Area

The study area is located within the experimental zone of the Rubber Research Institute of the Chinese Academy of Tropical Agricultural Sciences in Danzhou City, situated at the northwestern edge of Hainan Province, China (19°31'N, 109°29'E). The specific location is illustrated in Fig. 2(a). Danzhou lies in the East Asian tropics and experiences a pleasant tropical monsoon climate. The annual average temperature ranges from 22.5 °C to 25.6 °C, with January being the coldest month and July the hottest. Annual sunshine duration ranges from 1780 to 2600 h, and annual precipitation ranges from 900 to 2400 mm. The climate of Hainan Island is shaped by the combined influences of topography, monsoons, and tropical cyclones, resulting in distinct differences between eastern and western regions, mountainous and flat areas, as well as between wet and dry seasons. The study area encompasses approximately 15.2 ha of gently undulating terrain suitable

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TABLE I
SPECIFIC PARAMETERS OF CBI-LITE LIDAR MEASUREMENT CAMERA

Model	CBI Lite
Scanning Range	120 m (Reflectivity $\geq$ 60%)
Scanning Field of View	360°
Weight	1.28kg
Measurement Range	0–120 m
Measurement Accuracy	5mm (within 30m range)
Repeatability Accuracy	10mm
Scanning Frequency	5Hz 20Hz
Laser Wavelength	905nm
Echo Number	Dual Echo
Maximum Effective Rate	640,000 points/second
Camera	20 million pixels
Field of View	84°

for rubber plantation cultivation. Topographic analysis reveals elevation ranging from 125 to 158 m above sea level, with a total relief of 33 m across the site. Slope distribution shows that 65% of the area exhibits gradients less than 5°, 28% ranges between 5° and 15°, and 7% exceeds 15°, with a mean slope gradient of 6.8°. Located on the leeward side and adjacent to the Beibu Gulf, the region's favorable climate supports agricultural development, which has contributed to the continued expansion of rubber tree cultivation on Hainan Island.

B. Data Acquisition

During data acquisition, we used a DJIM300RTK UAV manufactured by DJI Innovations (Shenzhen, China), equipped with the CBI-Lite LiDAR system developed by Aurenav Technology (Chengdu, China), as shown in Fig. 2(b). The specifications of the LiDAR system are listed in Table I. The field experiment was conducted on March 5, 2023, and the sampling plot is shown in Fig. 2(c). To optimize the flight path of the UAV platform, a dual-grid flight strategy was implemented, as illustrated in Fig. 2(a). This approach ensured optimal LiDAR illumination on the tree trunks while capturing comprehensive structural details of surrounding trees. The UAV was flown at an altitude of 80 m, with a distance of approximately 50–60 m from the canopy. The average flight speed was 6 m/s, with a flight line spacing of 10 m and a lateral overlap rate of approximately 65%. The point cloud density of the sampled rubber plantation reached approximately 10 667 points/m^2. Fig. 2(d)–(g) shows the detailed images of tree species within the sampling area.

C. Data Preprocessing

To obtain high-quality rubber tree point cloud data, a systematic preprocessing workflow was adopted in this study. First, a statistical outlier removal algorithm [26] was applied to eliminate noise from the raw point cloud. Then, the cloth simulation filtering (CSF) algorithm [27] was used to separate ground and nonground points. Fig. 3 illustrates the complete workflow from raw point cloud data to height normalization. As shown in Fig. 3(a), the original 3-D point cloud contains both canopy and terrain surface information. After ground point separation [Fig. 3(b)], the extracted ground points were

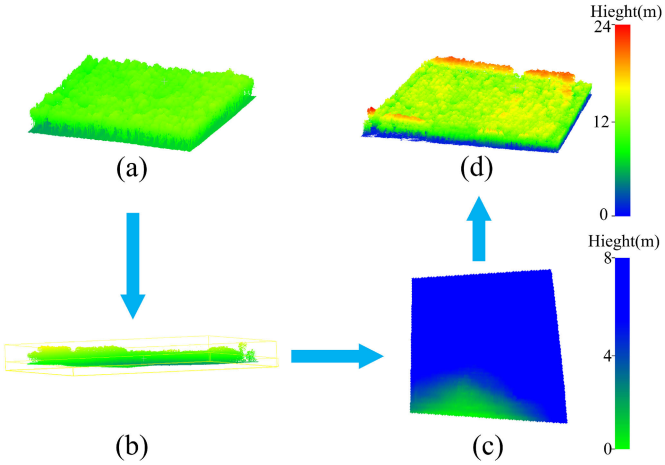


Fig. 3. Workflow of DEM generation and height normalization from LiDAR point cloud data. (a) Original 3-D point cloud with terrain surface. (b) Separated ground points after filtering. (c) Generated DEM raster (1-m resolution) showing elevation gradients from 0 to 8 m. (d) Height-normalized point cloud after terrain correction. The color scale represents height values in meters.

interpolated using the Delaunay triangulation algorithm to generate a digital elevation model (DEM) with 1-m spatial resolution [Fig. 3(c)]. The DEM clearly shows the elevation gradient across the study area, ranging from 0 to 8 m. Subsequently, height normalization was performed by computing the height difference between each point and the corresponding DEM elevation ($H_{\text{normalized}} = H_{\text{original}} - H_{\text{DEM}}$), resulting in terrain-corrected point cloud data [Fig. 3(d)]. Finally, considering the structural characteristics of rubber plantations, a combined approach using height thresholding and point density analysis [28] was applied to filter out understory vegetation, resulting in a relatively clean point cloud of rubber trees. Fig. 4(a) and (b) shows the CHM nadir view and profile view of a subset of the rubber plantation, clearly illustrating the spatial distribution pattern and vertical height gradient of the trees.

Individual tree segmentation and data annotation were performed by the research team led by Xiangjun Wang at the Rubber Research Institute of Hainan [29], [30], [31], [32] using the CloudCompare software. The annotation process utilized CHM data to facilitate tree identification. Starting from the CHM nadir view perspective shown in Fig. 4(a), initial region delineation was conducted based on canopy morphology and spacing between trees to isolate individual trees, as illustrated in Fig. 4(c). The profile view shown in Fig. 4(b) provides vertical structural information to assist in understanding tree height distribution and canopy layering. The segmented regions were then refined through multiangle inspection, including top, front, and side views, with particular attention paid to boundary determination in overlapping canopy areas. Precise segmentation was achieved by leveraging variations in point cloud density and geometric continuity. The final segmentation results were then annotated, as shown in Fig. 4(d). Each point in the point cloud was assigned two types of labels. The first type was a semantic label, where target rubber tree points were labeled as "1" (displayed in

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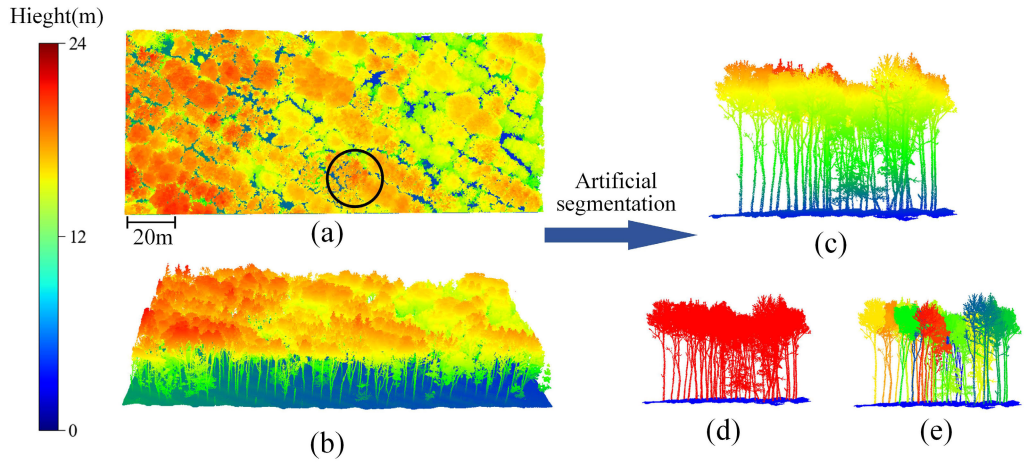


Fig. 4. Data preprocessing workflow. (a) Nadir view of the rubber tree point cloud showing CHM with normalized heights above ground level; (b) profile view of the rubber tree point cloud showing vertical structure; (c) manually segmented rubber tree blocks extracted from the raw point cloud; (d) semantic labeling result of segmented rubber trees, where red indicates target rubber trees and blue indicates nontree regions; and (e) instance labeling result, where different colors represent individual tree instances (labels "1," "2," and "3"), and blue represents nontree points (label "0").

red) and background points (including nontree and ground areas) were labeled as "0" (displayed in blue). The second type was an instance label, where each tree instance was assigned a unique identifier such as "1," "2," and "3," with each color representing a different instance, while background points remained labeled as "0." These two types of labels provide point-level manual annotations for both semantic class and tree instance ID. All annotation results were reviewed and validated by three experienced rubber tree experts, ensuring labeling accuracy exceeded 95%.

After the above preprocessing workflow, a total of 600 rubber trees were selected from thousands of segmented instances to construct the dataset. These sample trees were chosen to represent diverse canopy sizes, heights, and background complexities. The dataset was organized into 36 blocks, which were further divided into 23 training scenes, 8 validation scenes, and 5 testing scenes for effective model training and evaluation. The rubber tree point cloud dataset constructed in this study is partially available at https://doi.org/10.5281/zenodo.17149173 for research purposes. The complete dataset is available from the corresponding author if needed.

III. Methods

This study proposes a point cloud instance segmentation network, termed RtSegNet, for precise individual rubber tree segmentation and structural parameter extraction. The overall architecture of RtSegNet is illustrated in Fig. 5(a). The network comprises several key components, including 3D-Unet, TSILS, FSGC, and GR-DSA. First, the point cloud data are passed through a backbone network for feature extraction. The TSILS module, as illustrated in Fig. 5(b), performs tree-structure incremental learning-based sampling to generate high-dimensional feature representations. The network is divided into two parallel processing paths. The classification-confidence path performs semantic segmentation via the classification and confidence module, providing reliable foreground points for subsequent clustering. The instance segmentation path utilizes the FSGC module, as shown in Fig. 5(c), to model the spatial topology of seed points using a graph neural network and enhance feature discrimination through Fourier spectrum analysis. The GR-DSA module, as shown in Fig. 5(d), then applies a GRW-based mechanism to adaptively optimize clustering parameters. Finally, precise instance segmentation results are obtained through cluster filtering and 3-D nonmaximum suppression (NMS). All modules collaborate to enable efficient point cloud processing and accurate segmentation of individual rubber trees.

A. Tree-Structure Incremental Learning Sampling

To address the significant variation in rubber tree point cloud characteristics under different growth conditions and species, we propose a TSILS strategy, as illustrated in Fig. 5(b). TSILS adopts an incremental learning framework [33] that enables the model to progressively learn new rubber tree categories from a continuous data stream. Within this framework, a tree-structure-aware hierarchical sampling strategy is designed to construct a multilevel feature exemplar library, allowing the model to retain knowledge of previously learned classes. The proposed method consists of four core components: a decoupled model architecture, tree-structure hierarchical exemplar selection, a nearest mean classifier, and a knowledge distillation loss mechanism. The incremental learning process is represented as follows:

$$M_t = F\left(M_{t-1}, \mathcal{D}_t, \varepsilon_{t-1}\right) \tag{1}$$

where M_t denotes the updated model at time step t, M_{t-1} denotes the existing model at time step $t-1$, D_t denotes the newly arrived rubber tree point cloud dataset at time step t, ε_{t-1} denotes a representative set of samples (example set) selected and stored from the previous data, and F denotes an incremental learning update function that updates the model parameters based on the old model, new data, and historical examples. The new model is obtained through a learning process based on the old model, new data, and historical examples, rather than

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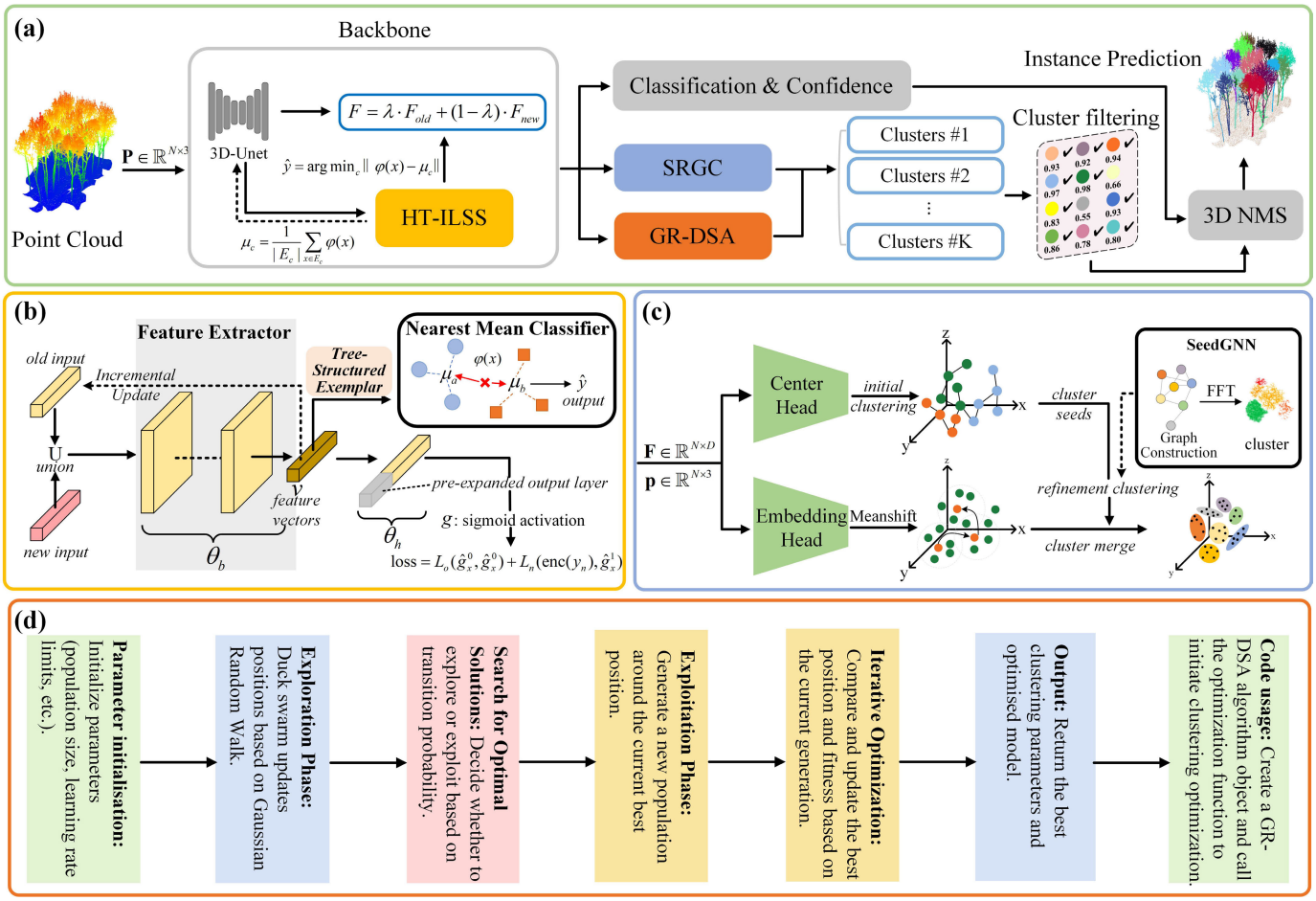


Fig. 5. Overview of the RtSegNet framework. (a) Overall architecture of RtSegNet. (b) TSILS. (c) FSGC. (d) GR-DSA.

being trained from scratch. The algorithm flow is as follows (see Algorithm 1):

$$v_m = \arg\max_{\|v\|=1} v^{\mathrm{T}} \operatorname{Cov}(P)\, v. \tag{2}$$

First, we design the decoupled model architecture. The model is decomposed into two key components, the feature extractor and the classifier

$$M = \psi \circ \varphi \tag{3}$$

where $\varphi : \mathcal{X} \to \mathcal{V}$ is a feature extractor that maps the input rubber tree point cloud data X to a d-dimensional feature space V, $\psi : \mathcal{V} \to \mathcal{Y}$ is a classifier that predicts class labels based on $v \in \mathcal{V}$ predicted category labels $y \in \mathcal{Y}$, and $\circ$ denotes a function composite operation. The advantage of this decoupled design is that the class prototype is automatically updated when the parameters of the feature extractor change, ensuring that the classifier is robust to changes in the feature representation.

Second, considering the unique geometric structure characteristics of rubber trees, we designed a hierarchical example selection algorithm based on tree structure. For each tree's point cloud data $P = \{p_1, p_2, \ldots, p_n\}$, where $p_i \in \mathbb{R}^3$ denotes the 3-D coordinates of the ith point, the specific selection process is shown in Fig. 6, and we compute the direction vector of the main trunk of the rubber tree by principal component analysis (PCA) [34]

$$v_m = \arg\max_{\|v\|=1} v^{\mathrm{T}} \operatorname{Cov}(P)\, v \tag{4}$$

where $v_m \in \mathbb{R}^3$ is the unit direction vector of the trunk axis, $\operatorname{Cov}(P)$ is the 3×3 covariance matrix of the point cloud data, and the constraints $\|v\| = 1$ ensure that the solution is a unit vector. The solution of the optimization problem corresponds to the eigenvector corresponding to the largest eigenvalue of the covariance matrix, reflecting the main direction of the point cloud data, i.e., the trunk direction of the rubber tree.

Based on the trunk axis, we vertically divide the tree into L layers, with the height range of each layer defined as

$$L_i = \left\{ p \in P \,\middle|\, h_{\min} + \frac{(i-1)\,H}{L} \le h(p) < h_{\min} + \frac{iH}{L} \right\} \tag{5}$$

$$H = \max_{p \in P} h(p) - h_{\min} \tag{6}$$

where L_i denotes the set of points in layer i, $h_{\min} = \min_{p \in P} h(p)$ is the minimum height of the tree, and $H = \max_{p \in P} h(p) - h_{\min}$ is the total height of the tree. The height function $h(p)$ calculates the coordinates of the projection of point p on the trunk axis, and $\bar{p} = 1/n \sum_{i=1}^{n} p_i$ is the center of mass of the point cloud.

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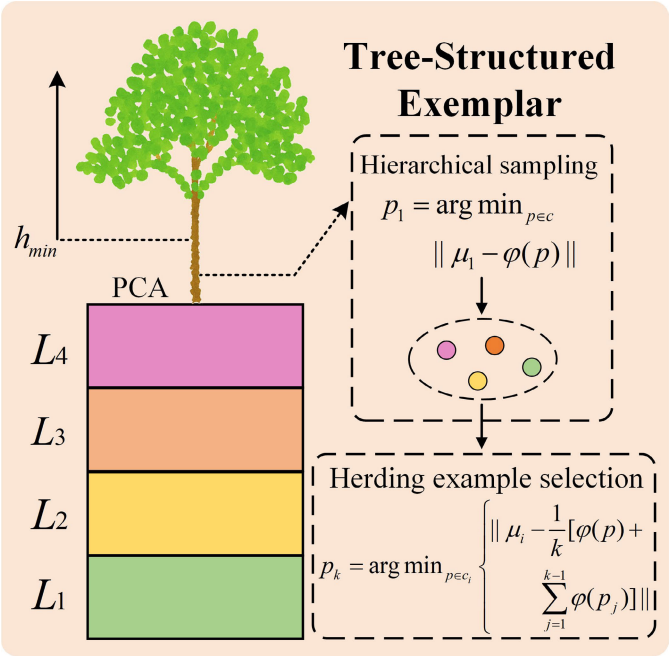


Fig. 6. Workflow of the tree-structure exemplar construction process. The main trunk axis is extracted using PCA, and the tree is vertically divided into multiple hierarchical levels (L1–L4) for layered sampling. In each level, representative samples are selected using the herding strategy to construct a multilevel structural exemplar library.

Within each layer, we use a herding-based example selection mechanism to iteratively select the samples that bring the mean of the example set closest to the mean of the full volume of data within the layer

$$p_k = \arg\min_{p \in L_i} \left\| \mu_i - \frac{1}{k} \left[\varphi(p) + \sum_{j=1}^{k-1} \varphi(p_j) \right] \right\| \tag{7}$$

where p_k is the kth selected sample, $\mu_i = 1/|L_i| \sum_{p \in L_i} \varphi(p)$ is the mean feature vector of the ith layer, $\varphi(p)$ is the feature vector of point p, and $\sum_{j=1}^{k-1} \varphi(p_j)$ is the sum of the feature vectors of the $k-1$ samples that have been selected. This prioritization construction ensures that the mean feature vector of any subset from the first example is a good approximation to the mean vector of that layer. For the high intraclass variability of the rubber tree morphology, we employ the nearest mean classifier [35], which differs from other incremental classification strategies such as iCaRL by using direct prototype-based classification rather than exemplar replay mechanisms. This approach provides computational efficiency and interpretability while avoiding the memory overhead of maintaining historical sample sets

$$y^* = \arg\min_{i=1,\ldots,t} \| \varphi(x) - \mu_i \| \tag{8}$$

where y^* is the predicted class label, $\mu_i = 1/|P_i| \sum_{p \in P_i} \varphi(p)$ is the feature mean of class i, P_i is the set of example points of class i, and $\| \cdot \|$ denotes the Euclidean distance. This classification strategy does not rely on decoupled weight vectors, and the class prototypes are automatically updated when the feature representations change, thus making the classifier robust to changes in feature representations.

Finally, to prevent forgetting existing knowledge when learning new rubber tree categories, we use an improved knowledge distillation loss function

$$L(\theta) = -\sum_{(x_i, y_i) \in \mathcal{D}} [L_{\text{new}} + L_{\text{old}}] \tag{9}$$

where θ denotes all parameters of the model, D is the augmented training set containing the old category examples and the new category data, and (x_i, y_i) is the ith training sample and its corresponding label.

The new category loss term is defined as

$$L_{\text{new}} = \sum_{y=s}^{t} \left[\delta_{y=y_i} \log g_y(x_i) + \delta_{y \neq y_i} \log \left(1 - g_y(x_i)\right) \right] \tag{10}$$

where s is the starting index of the new category; t is the current total number of categories; $\delta_{y=y_i}$ is the Kronecker δ function, which is 1 when $y = y_i$ and 0 otherwise; and $g_y(x_i)$ is the model's predicted probability that a sample x_i belongs to category y, using a sigmoid activation function.

The old category distillation loss term is defined as

$$L_{\text{old}} = \sum_{y=1}^{s-1} \left[q_i^y \log g_y(x_i) + \left(1 - q_i^y\right) \log \left(1 - g_y(x_i)\right) \right] \tag{11}$$

where q_i^y is the predicted probability of the old model that sample x_i belongs to category y. This loss ensures that the output of the new model on the old categories is consistent with the old model.

TSILS effectively addresses interspecies variability in rubber trees through four core technical innovations. The tree-structure hierarchical sampling mechanism captures key morphological features, while the herding-based exemplar selection strategy maximizes information retention under limited memory constraints. A nearest mean classifier provides robust classification decisions, and the knowledge distillation technique enables balanced integration of new and previously learned knowledge. These synergistic components ensure that the model maintains efficient incremental learning performance when faced with species diversity. The effectiveness of TSILS is demonstrated through experiments presented in Section IV-C1

B. Fourier Spectrum-Enhanced Spatial Graph Clustering

To address the critical challenge of overlapping canopies in rubber tree point cloud instance segmentation, we propose an FSGC method, as illustrated in Fig. 5(c). FSGC adopts a dual-branch parallel processing architecture, in which two heads collaboratively mitigate the impact of canopy overlap. The center head branch starts with initial cluster centers generated by a region growing algorithm, and constructs a seed point graph structure. This graph is refined using a graph neural network [36] for spatial relationship modeling. To further resolve oversegmented clusters, Fourier spectrum analysis is applied to merge clusters based on topological coherence, yielding spatial topology-aware segmentation results. The embedding head branch, on the other hand, uses a specially designed embedding network to map raw point features into a low-dimensional representation space. In this space, mean shift

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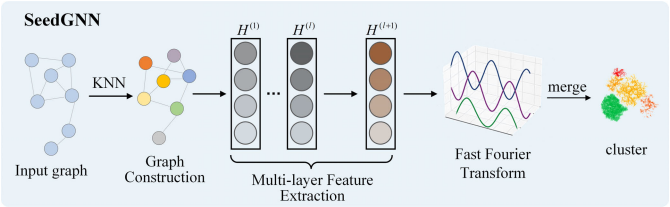


Fig. 7. Workflow of SeedGNN. The method constructs the initial input graph with seed points and establishes the spatial relationship graph by K-nearest neighbor algorithm; subsequently, a graph neural network is used for multilayer feature extraction to obtain the node feature matrix. The final features are subjected to FFT and clustered and merged based on spectral similarity to obtain the results.

clustering is applied to perform density-based segmentation, providing complementary information to the center head. The clustering results from the two branches are subsequently integrated to accurately distinguish individual tree instances in overlapping canopy regions.

First, we denote the input point cloud as $\{(p_i, F_i)\}_{i=1}^{N}$, where N represents the number of points in the point cloud, $F_i \in \mathbb{R}^D$ represents the 3-D coordinates of point i, $F_i \in \mathbb{R}^D$ is the feature vector of point i, and D is the feature dimension. In the center head branch, we apply a region growing algorithm to the point cloud to obtain an initial oversegmentation, resulting in coarse clustering outputs

$$C = f_{\text{region}}(p, L, r) \tag{12}$$

where $p \in \mathbb{R}^{N\times 3}$ is the set of point cloud coordinates, $L \in \{0,1\}^N$ is the semantic label (indicating whether each point belongs to a rubber tree or not), r is the search radius parameter for region growing, f_{region} denotes the region growing algorithm, $C = \{C_1, C_2, \ldots, C_M\}$ is the initial clustering result, and M is the number of clusters.

The region growing algorithm forms initial clusters by selecting seed points with high confidence and iteratively adding neighboring points to the current region while satisfying the distance and feature similarity conditions. For each initial cluster C_i, we compute its spatial center and feature representation

$$\S_i = \frac{1}{|C_i|}\sum_{j\in C_i} p_j, F_i^s = \frac{1}{|C_i|}\sum_{j\in C_i} F_j \tag{13}$$

where $S_i \in \mathbb{R}^3$ denotes the spatial center coordinates of the ith cluster, $|C_i|$ denotes the number of points contained in cluster C_i, and $F_i^s \in \mathbb{R}^D$ denotes the average features of the ith cluster. S_i denotes the geometric center of the cluster, and F_i^s incorporates the geometric and neighborhood statistical features of the points in the region.

Then, we design the graph neural network `SeedGNN` to refine the spatial relationship perception of the seed features as shown in Fig. 7. Enhance the characterization of the seed points, and construct the spatial relationship graph $G = (V, E)$ between the seed points, where $V = \{1, 2, \ldots, M\}$ is the set of seed points and E is the set of edges, which is defined by the K-nearest neighbor algorithm

$$E = \left\{(i,j) \mid d\left(S_i, S_j\right) < \epsilon_k\left(S_i\right)\right\} \tag{14}$$

where $d(S_i, S_j) = \|S_i - S_j\|^2$ is the Euclidean distance between seed points and $\epsilon_k(S_i)$ is the distance threshold for the kth nearest neighbor of point S_i.

Feature refinement based on SeedGNN can be expressed as follows:

$$H^{(l+1)} = \sigma\left(D^{-\frac{1}{2}}\mathrm{A}D^{-\frac{1}{2}}H^{(l)}W^{(l)}\right) \tag{15}$$

where $H^{(l)} \in \mathbb{R}^{M\times d_l}$ is the node feature matrix of layer l; the initial feature $H^{(0)} = F^s$ (i.e., the matrix consisting of the average features of all seed points); $A \in \mathbb{R}^{M\times M}$ is the adjacency matrix, in which $A_{\text{ij}} = 1$ if $(i,j) \in E$, otherwise $A_{\text{ij}} = 0$; $D \in \mathbb{R}^{M\times M}$ is the degree matrix, which is a diagonal matrix with the diagonal element $D_{\text{ii}} = \sum_j A_{\text{ij}}$ denoting the degree of node i; $W^{(l)} \in \mathbb{R}^{d_l\times d_{l+1}}$ is the learnable weight matrix of layer l; and σ is the ReLU nonlinear activation function. $D^{-1/2}\mathrm{A}D^{-1/2}$ is the normalization operation on the adjacency matrix to ensure the numerical stability of the feature propagation process. After L-layer graph convolution, we obtain the final seed point features $H^{(L)} \in \mathbb{R}^{M\times d_L}$, which contain rich spatial relationship information.

Subsequently, we designed a clustering and merging strategy for spectral analysis. We perform a fast Fourier transform (FFT) [37] on the refined features to obtain their spectral representation

$$F_s = T\left(H^{(L)}\right) \tag{16}$$

where T denotes the Fourier transform operation and $F_s \in \mathbb{R}^{M\times d}$ is the spectral feature matrix. The spectral features extract the frequency components of the feature vectors by the Fourier transform, focusing on the periodicity and oscillatory patterns of feature changes.

Based on the spectral features, we calculate the similarity matrix between the seed points $S \in \mathbb{R}^{M\times M}$

$$S_{\text{ij}} = \frac{F_s^i \cdot F_s^j}{\|F_s^i\| \cdot \|F_s^j\|} \tag{17}$$

where F_s^i denotes the spectral eigenvector of the ith seed point, $\cdot$ denotes the vector inner product operation, and $\|\cdot\|$ denotes the Euclidean norm of the vector. This cosine similarity measure can effectively evaluate the degree of similarity between two seed points in the spectral space, ignoring the absolute size of the vectors and focusing only on the directional similarity. When the similarity S_{ij} exceeds a preset threshold τ, we merge the corresponding clusters

$$C_m = \left\{C_i \cup C_j \mid S_{\text{ij}} > \tau \quad \forall i, j \in \{1, 2, \ldots, M\}, i \neq j\right\} \tag{18}$$

where C_m denotes the merged clustering result and τ is the similarity threshold, which is usually determined based on the experimental results on the validation set. The merging strategy based on spectral similarity is capable of mitigating the oversegmentation caused by the initial region growing step, by reassociating clusters that originally belong to the same tree but were incorrectly split.

Meanwhile, the embedding head branch provides complementary segmentation information to map the point features to the low-dimensional space

$$Z = \Phi(F) \tag{19}$$

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where $\Phi : \mathbb{R}^{N\times D} \rightarrow \mathbb{R}^{N\times d_Z}$ denotes the embedding network, $Z \in \mathbb{R}^{N\times d_Z}$ is a low-dimensional embedded feature, and d_Z is usually smaller than the original feature dimension D.

The embedding network uses a two-layer MLP structure to map the feature dimensions of the backbone output into a d_Z-dimensional space by learning a discriminative representation at the point level. A mean shift algorithm is then applied in the embedding space for clustering

$$C_s = \mathrm{MS}\,(Z, \gamma) \tag{20}$$

where MS denotes the mean shift algorithm, γ is a bandwidth parameter that controls the granularity of clustering, and C_s is the result of mean shift clustering. The mean shift algorithm is a density-based clustering method that finds patterns in the distribution of data by iteratively moving data points to regions of maximum local density.

Finally, the multisource clustering results are integrated to produce the final instance segmentation outcome

$$C_f = M\,(C_m, C_{\mathrm{ms}}) \tag{21}$$

where M is the fusion function, which is fused using the voting mechanism of overlap degree. Specifically, for each point, it is first determined to which cluster it belongs to in C_m and C_{ms}, respectively, and the label is directly adopted when the two branches assign the point to the corresponding instance of the same tree; when the assignment results are inconsistent, the closer cluster is selected as the final label by calculating the distance between the point and the centers of the two candidate clusters. For points with fuzzy boundaries, the weights of the two branches are dynamically adjusted according to the spatial density characteristics of the points.

Overall, the key of FSGC to solve the tree canopy-overlapping problem lies in the spatial relationship perception of graph neural network and Fourier spectrum analysis. FSGC learns the distinguishing features of different trees in the overlapping region by explicitly modeling the seed point relationship through the graph structure. Meanwhile, spectral analysis further enhances the feature differentiation by revealing the frequency patterns in the time-domain features, effectively capturing the subtle differences in the shape and structure of the tree canopy. The effectiveness experiment of FSGC is presented in Section IV-C2.

C. Gaussian Random Duck Swarm Algorithm

To address the variation in rubber tree sizes, careful selection of clustering bandwidth parameters is essential. To this end, we adopt the duck swarm algorithm (DSA) [38] to search for optimal parameter values. The DSA consists of two main phases: exploration and exploitation. However, relying solely on these two phases may lead to premature convergence or deviation from the global optimum. To overcome this limitation, we integrate GRWs into the DSA and propose an enhanced version termed the GR-DSA.

The DSA initially uses the following expression to randomly generate initial positions in a D-dimensional search space:

$$X_i = L_b + (U_b - L_b) \cdot o \tag{22}$$

where X_i denotes the position vector representing the ith duck within the population; $(i = 1, 2, 3, \ldots, N)$ is the total count of distinct populations; N is the size of each population group; L_b and U_b denote the defined search range for the learning rate's upper and lower limits, respectively; and o is a matrix composed of uniformly distributed random values in the interval $[0, 1]$.

Once the queuing phase concludes, namely when the ducks reach regions abundant in food resources, they begin to scatter and initiate independent food-seeking behavior, transitioning into the exploratory stage. To mitigate the issue of the duck flocking algorithm becoming trapped in local optima during food search, we incorporate a stochastic strategy known as GRW [39] in the exploration process. The GRW strategy, commonly employed to model stochastic trajectories in low-dimensional spaces, models movements in accordance with the specifics of a Gaussian normal distribution. In a D-dimensional setting, the motion at each iteration is represented by

$$X_{t+1} = X_t + \sigma \cdot N\,(0, I) \tag{23}$$

where X_t represents the current position as a D-dimensional vector, σ is a scalar or D-dimensional vector indicating the step length in each corresponding dimension, and $N(0, I)$ denotes a D-dimensional normally distributed random variable with mean 0 and covariance matrix I

$$X_i^{(t+1)} = \begin{cases} X_i^t + \mu X_i^t \mathrm{sign}\,(r - 0.5) + \sigma N\,(0, I) \\ \quad \text{if } P > \mathrm{rand} \\ X_i^t + CF_1\left(X_l^t - X_i^t\right) + CF_2\left(X_j^t - X_i^t\right) \\ \quad \text{if } P \leq \mathrm{rand} \end{cases} \tag{24}$$

where $\mathrm{sign}(r-0.5)$ affects the process of searching for food and can be set to -1 or 1, denoting the control parameter of global search. P is the search transition probability in the exploration phase, and CF_1 and CF_2 denote the cooperation and competition coefficients between ducks in the search phase, respectively. X_l^t denotes the optimal value of the position of all ducks after the tth iteration, and X_j^t denotes the ducks searching for food in the vicinity of X_i^t in the tth iteration. $\sigma \cdot N(0, I)$ represents the stochastic perturbation introduced using a Gaussian distribution. The parameter μ involved in the formulation is determined by the following expression:

$$\mu = K \cdot \left(1 - \frac{t}{t_{\max}}\right). \tag{25}$$

Specifically, K is calculated as follows:

$$K = \sin\,(2 \cdot \mathrm{rand}) + 1. \tag{26}$$

When the ducks find enough food, they transition into the exploitation stage. During the development stage, a new set of candidate solutions is stochastically generated based on Gaussian perturbations guided by the leading individual of the current population. We define the development phase of GR-DSA using the following equation:

$$X_{t+1} = \mathrm{Gaussian}\left(X_{\mathrm{best}}^t, \sigma\right) + \left(r_3 \cdot X_{\mathrm{best}}^t\right) - r_4 \cdot X_i^t \tag{27}$$

where $\mathrm{Gaussian}(X_{\mathrm{best}}^t, \sigma)$ refers to a stochastic value drawn from a distribution centered at X_{best}^t with variance σ, X_i^t is the

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Algorithm 1 TSILS

Require: Point cloud $P = \{p_1, p_2, \ldots, p_n\}$, layer number L, exemplar number m
Ensure: Exemplar set $\mathcal{E} = \{e_1, e_2, \ldots, e_m\}$ HierarchicalSelection$P$, L, m

$\mathbf{v}_{\text{main}} \leftarrow \text{PrincipalComponent}(P)$
for each $i = 1$ to L **do**
  $L_i \leftarrow \left\{p \in P \mid h_{\min} + \frac{(i-1)H}{L} \le h(p) < h_{\min} + \frac{iH}{L}\right\}$
end for
$\mathcal{E} \leftarrow \emptyset$
for each layer L_i **do**
  $\mu_i \leftarrow \frac{1}{|L_i|}\sum_{p\in L_i}\varphi(p)$
  for each $k = 1$ to $\lceil \frac{m}{L} \rceil$ **do**
    $p_k \leftarrow \arg\min_{p\in L_i} |\mu_i - \frac{1}{k}\left[\varphi(p) + \sum_{j=1}^{k-1}\varphi(p_j)\right]|$
    $\mathcal{E} \leftarrow \mathcal{E} \cup \{p_k\}$
  end for
end for
return $\mathcal{E}$

TABLE II
GR-DSA PARAMETER CONFIGURATION

Parameter	Value	Selection Method
Population Size	20	Validation experiments
CF1 (Cooperation)	0.8	Grid search
CF2 (Competition)	0.4	Grid search
P (Transition probability)	0.7	Sensitivity analysis
σ (Step size)	Adaptive	Eq. (28)
Max iterations	50	Convergence analysis
Bandwidth bounds	[0.4, 1.4]	Domain knowledge

optimal individual in the tth iteration, r_3 and r_4 are uniformly distributed random variables within the $[0, 1]$ range, and the variance σ is computed via the following formula:

$$\sigma = \frac{\log t}{t} \cdot \left(X_i^t - X_{\text{best}}^t\right). \tag{28}$$

The optimized clustering parameters obtained through the GR-DSA algorithm facilitate more effective model training. As evidenced in (28), it is observable that generating new populations near the population leader by leveraging the current leading individual significantly accelerates the convergence speed of the algorithm. The variance σ is an adaptive value. This formulation assigns a larger σ at the initial stages, thus expanding the search scope and enhancing the algorithm's ability to escape local optima. As the algorithm converges, σ diminishes, reducing fluctuations and thereby promoting more precise local refinement. A full depiction of the GR-DSA algorithm's pseudo-code structure is provided in Algorithm 2.

The GR-DSA implementation requires careful parameter configuration to balance exploration capability with computational efficiency. Table II summarizes the key parameters and their selection strategies.

The population size of 20 particles provides sufficient diversity for bandwidth optimization while maintaining computational tractability. Cooperation coefficient CF1 = 0.8 promotes effective information sharing among particles, while competition coefficient CF2 = 0.4 preserves population diversity to prevent premature convergence. The transition probability P = 0.7 balances exploration and exploitation

Algorithm 2 GR-DSA

Require: Population size N, dimensions d, bounds lb, ub, max iterations T
Ensure: Optimal parameters θ^*

for each duck $i = 1$ to N **do**
  $X_i \leftarrow lb + rand(0,1) \cdot (ub - lb)$
end for
for $t = 1$ to T_1 **do**
  for each duck i **do**
    $X_i^{t+1} \leftarrow X_i^t + \alpha \cdot \mathcal{N}(0, \sigma^2) + a \cdot (X_{best}^t - X_i^t) + b \cdot (X_j^t - X_i^t)$
    $X_i^{t+1} \leftarrow clip(X_i^{t+1}, lb, ub)$
  end for
end for
for $t = T_1 + 1$ to T **do**
  for each duck i **do**
    $r_1 \sim U(0,1), r_2 \sim U(0,1)$
    $\sigma^2 \leftarrow \sigma_0^2 / t$
    $X_i^{t+1} \leftarrow X_{best}^t + r_1 \cdot \mathcal{N}(0, \sigma^2) + r_2 \cdot (X_i^t - X_{best}^t)$
  end for
end for
$\theta^* \leftarrow X_{best}^T$
return θ^*

TABLE III
HARDWARE AND SOFTWARE PARAMETERS

Taining Parameter	Setting
Activation Functionn	Sigmoid
Batch size	16
Number epoch	150
Optimizer	Adam
Learning Rate Scheduler	ExponentialLR
BatchNorm Scheduler	BNMomentumScheduler

phases, determined through sensitivity analysis on validation data. The adaptive step size σ follows the decay function in (28), beginning with broad exploration and gradually focusing on local refinement. Bandwidth bounds [0.4, 1.4] encompass practical clustering parameter ranges based on mean shift clustering requirements for point cloud data. The effectiveness experiment of GR-DSA is presented in Section IV-C3.

IV. EXPERIMENTS

This section validates the effectiveness of RtSegNet through a series of experiments. First, ablation studies and effectiveness experiments are conducted to assess the contributions of the FSGC, CSCA modules, and GR-DSA. Next, RtSegNet is trained and validated, demonstrating its performance advantages over state-of-the-art point cloud segmentation networks. Subsequently, the model is tested on the For-instance and NIBIO MLS public datasets to compare its generalization capability. Finally, structural parameters of rubber trees are extracted from the segmentation results, further showcasing the practical value of the network.

A. Experimental Environment and Parameter Setting

The experimental parameters are defined as follows: the CPU used is Intel[1] Xeon[1] Platinum 8255C CPU @ 2.50 GHz,

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the GPU is NVIDIA GeForce RTX 3090, cuda version is 11.1, the system environment is Ubuntu 18.04, the Python environment is 3.8, the framework used is Pytorch 1.8.1, the RAM is 43.0 GB, and the number of training rounds is 150. The training parameters are shown in Table III.

B. Evaluation Metrics

To evaluate the segmentation performance of different models, we adopt a point-wise intersection over union (IoU)-based instance matching strategy. A predicted tree (PT) is considered a true positive (TP) if it has a corresponding ground truth (GT) instance and its IoU exceeds 0.5. Based on this criterion, we obtain the counts of TPs, false positives (FPs), and false negatives (FNs), which are used to compute commonly used instance-level segmentation metrics, including

$$\text{Iacc} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{29}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{30}$$

$$\text{mIoU} = \frac{1}{N}\sum_{i=1}^{N} \text{IoU}_i \tag{31}$$

$$F\text{-score} = \frac{2 \times \text{mPre} \times \text{mRec}}{\text{mPre} + \text{mRec}} \tag{32}$$

$$\text{mPre} = \frac{1}{N}\sum_{i=1}^{N} \frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i} \tag{33}$$

$$\text{mRec} = \frac{1}{N}\sum_{i=1}^{N} \frac{\text{TP}_i}{\text{TP}_i + \text{FN}_i}. \tag{34}$$

To comprehensively evaluate the performance stability of the model under different IoU thresholds, we introduce the mean average precision (mAP), which is calculated as the mean of the average precision values across multiple IoU thresholds

$$\text{mAP} = \frac{1}{|T|}\sum_{t \in T} \text{AP}_t \tag{35}$$

where T is a set of IoU thresholds and AP_t denotes the average precision computed at threshold t. The mAP metric provides a more comprehensive evaluation of the model's overall performance under varying levels of overlap requirements, thus avoiding the potential bias introduced by relying on a single threshold.

Verifying the existence of tree instances is only a preliminary step in the assessment; a more in-depth analysis requires assessing the consistency of the predicted instances with the actual boundaries. Thus, we introduce the coverage metric (Cov), which quantifies the consistency with the GT instance boundaries. Denoting the set of GT trees as $\mathcal{T}_{\text{gt}}$ and the set of PTs as $\mathcal{T}_{\text{pred}}$, for each GT tree we find the prediction instance with the highest IoU score, formally

$$\max \text{IoU}\left(l_i^{\text{gt}}\right) = \max\left(\text{IoU}\left(l_i^{\text{gt}}, l_j^{\text{pre}}\right)\right). \tag{36}$$

The coverage is then defined as the average over all reference (GT) trees

$$\text{Cov} = \frac{1}{N_{\text{gt}}}\sum_{i=1}^{N_{\text{gt}}} \max \text{IoU}\left(l_i^{\text{gt}}\right). \tag{37}$$

TABLE IV
PERFORMANCE COMPARISON OF INCREMENTAL LEARNING INTEGRATION AT DIFFERENT STAGES

Method	Cov	mPre	mRec	F-score	mAP	Iacc	mIoU
Clustering stage	72.47	90.65	80.71	82.38	73.51	87.9	68.62
Feature extraction stage	73.9	90.26	80.84	83.5	75.91	87.52	69.14

TABLE V
COMPARISON OF INCREMENTAL LEARNING INTEGRATION STAGES

Method	Cov	mPre	mRec	F-score	mAP	Iacc	mIoU
iCaRL	71.67	85.53	78.49	82.46	73.13	85.48	67.97
Category Enhancement	74.28	91.45	80.48	84.91	74.79	88.02	70.36
TSILS(ours)	**75.85**	**91.02**	**80.84**	**85.27**	**75.8**	**90**	**72.42**

In addition, we also considered the root-mean-square error (RMSE) of tree height (H: above ground level), calculated as the difference between the highest and lowest point of each tree instance. The motivation for including the tree height RMSE is that it represents a complementary metric to the detection rate and F-score scores, capturing the ability of the model to segment the entire length of the tree. The RMSE is calculated according to the following equation:

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(H_{\text{GT},i} - H_{\text{Pred},i}\right)^2}. \tag{38}$$

C. Module Effectiveness Experiment

In this section, the three core modules of the RtSegNet model, TSILS, FSGC, and GR-DSA, are analyzed in depth. Under the same experimental setting, the effectiveness verification was used to demonstrate the contribution of each module to the overall performance of the rubber tree instance segmentation task.

1) Effectiveness Experiment of TSILS: To further validate the effectiveness of the TSILS module, we designed two sets of experiments to analyze its underlying mechanism. As shown in Table IV, the module ensemble position experiment reveals an important mechanism: the ensemble effect of TSILS in the feature extraction stage is significantly better than that in the clustering stage, and the mAP is increased to 75.91%. The essence of this phenomenon lies in the stability of the feature space in the process of knowledge distillation. Incremental learning is introduced in the feature extraction stage, so that the retention of old knowledge is realized through the constraints at the feature level, and the damaging impact of classifier weight adjustment on the learned representations is avoided. However, the intervention in the clustering stage faces the problem that the feature representation has been solidified, and the flexibility of incremental update is limited.

Based on the optimal integration point at the feature extraction stage, TSILS achieves significant performance improvements. As shown in Table V, compared with conventional incremental learning methods, TSILS improves the mean IoU (mIoU) to 72.42% (mIoU: +4.45%) and reaches an F-score of 85.27%. This performance gain primarily stems from TSILS overcoming the limitations of randomness in

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TABLE VI
COMPARISON OF FSGC WITH 2-D CLUSTERING METHODS

Method	Cov	mPre	mRec	F-score	mAP	Iacc	mIoU
MiniBatchKMeans	31.7	76.73	35.27	47.87	31.38	76.6	31.29
GMM	32.79	85.22	38.41	52.39	35.93	85.07	32.6
Birch	58.45	84.24	44.83	58.06	41.33	84.25	37.78
FSGC (Ours)	**76.09**	**92.47**	**79.89**	**84.79**	**74.93**	**90.65**	**72.59**

TABLE VII
PERFORMANCE COMPARISON OF FSGC WITH POINT CLOUD CLUSTERING METHODS

Method	Cov	mPre	mRec	F-score	mAP	Iacc	mIoU
Region Growing	68.4	85.7	72.4	79.2	59.18	87.1	64.2
HDBSCAN	64.5	83.3	70.2	79.1	64.17	86.5	64.6
Meanshift	72.1	84.3	76.1	80.9	67.8	87.4	65.9
Meanshift+HDBSCAN	74.6	88.5	79.2	82.5	71.42	90.2	69.6
FSGC (Ours)	**76.09**	**92.47**	**79.89**	**84.79**	**74.93**	**90.65**	**72.59**

TABLE VIII
EFFECTIVENESS EVALUATION OF THE GR-DSA ALGORITHM

Method	Cov	mPre	mRec	F-score	mAP	Iacc	mIoU
PSO	69.3	83.3	76.1	82.3	69.36	82.9	60.1
BA	71.8	85.2	77.4	82.4	67.56	88.1	62.4
DE	74.94	87.72	79.36	83.08	70.59	88.95	67.04
GR-DSA	**75.95**	**92.47**	**79.89**	**84.79**	**74.93**	**90.65**	**72.59**

traditional exemplar selection strategies. Traditional incremental learning typically constructs exemplar sets via random sampling or simple similarity-based selection, which overlooks the hierarchical geometric characteristics of rubber tree point cloud data. Rubber trees exhibit a distinct morphological gradient from the root to the crown, with noticeable variations in point distribution patterns, density, and geometric complexity across vertical layers. TSILS identifies the main trunk axis and performs vertical stratified sampling accordingly, ensuring that the exemplar set captures the full structural spectrum of each tree. More importantly, the integration of the herding algorithm enables exemplar selection in each layer to follow the principle of minimizing the discrepancy between the exemplar mean and the full-layer data mean. This constraint ensures that the limited exemplar set provides an optimal approximation of the original data distribution.

2) Effectiveness Experiment of FSGC: To verify the effectiveness of the FSGC module, we conducted comprehensive comparisons against both 2-D baseline methods and existing 3-D clustering approaches. We evaluated FSGC against traditional 2-D clustering methods applied to projected point cloud data, including MiniBatchKMeans, Birch clustering, and GMMs. As shown in Table VI, these 2-D approaches demonstrated significant limitations. MiniBatchKMeans achieved only 31.29% mIoU due to its inability to handle complex crown shapes in 2-D projections. Birch clustering showed moderate improvement through hierarchical processing but suffered from substantial information loss during 3-D-to-2-D projection, achieving 37.78% mIoU. GMM's probabilistic clustering approach in 2-D space lacks access to the 3-D structural information critical for distinguishing overlapping tree crowns, resulting in similar performance limitations with 32.6% mIoU. We further compared FSGC with several 3-D clustering methods, including region growing [40], HDBSCAN [41], mean shift [42], and a combined mean shift + HDBSCAN approach [43], as shown in Table VII. The region growing method offers advantages such as simplicity and intuitive parameter tuning. However, since it relies solely on Euclidean distance for point cloud clustering, it struggles to accurately separate points belonging to different tree instances that are closely spaced in crown-overlapping regions, resulting in blurred boundaries and mis-segmentation. The HDBSCAN algorithm, which performs density-based clustering, can handle clusters of arbitrary shapes. Nevertheless, in the presence of overlapping rubber tree crowns with highly similar geometric features, its density-based criteria fail to define clear instance boundaries. Although the mean shift + HDBSCAN hybrid method combines the strengths of both strategies and improves performance to some extent, it still cannot fundamentally address the modeling of complex 3-D nonlinear spatial topologies. In contrast, FSGC constructs a seed point graph structure to transform implicit spatial relationships into explicit topological connections. By leveraging the message-passing mechanism of graph neural networks, each seed point aggregates multihop neighborhood information, thereby capturing the global spatial context and enhancing sensitivity to subtle structural differences in overlapping regions. Additionally, the spectral clustering merge strategy transforms spatial-domain features into the frequency domain, offering extra discriminative cues for distinguishing crown regions that are morphologically similar but belong to different instances. In contrast to both 2-D and traditional 3-D approaches, FSGC achieved superior performance (72.59% mIoU), demonstrating the necessity of 3-D spatial modeling combined with graph-based topological reasoning for accurate tree crown segmentation.

3) Effectiveness Experiment of GR-DSA: To evaluate the performance of the GR-DSA algorithm, we compared it with three optimization algorithms: PSO [44], BA [45], and DE [46]. These algorithms were applied to optimize the clustering parameters for rubber trees, and the experimental results are shown in Table VIII. The PSO algorithm searches for optimal parameters through information sharing among particles, but it is prone to local optima in multimodal functions. The BA algorithm, which simulates bat echolocation behavior, has a certain global search ability but converges slowly and lacks adaptability to variations in rubber tree scale. The DE algorithm maintains population diversity through differential mutation, yet its fixed-parameter update mechanism struggles to handle spatial variations in point cloud density. In comparison, GR-DSA enhances the original DSA by integrating a GRW mechanism that adaptively adjusts search behavior via variance control. The algorithm performs global exploration with a larger variance in the early stage and gradually reduces variance for local convergence. Moreover, it dynamically adjusts clustering parameters based on the density distribution of rubber tree point clouds, transforming global optimization into a locally adaptive process. Experimental results demonstrate that GR-DSA outperforms traditional

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TABLE IX
RESULTS OF ABLATION EXPERIMENTS

Methods			Evaluation Metrics							
TSILS	FSGC	GR-DSA	Cov(%)	mPre(%)	mRec(%)	F-score(%)	mAP(%)	Iacc(%)	mIoU(%)	Training Time(h)
			73.11	87.92	76.21	80.41	72.72	86.41	66.49	46.91
✓			75.85	91.02	80.84	85.27	75.8	90	72.42	**17.44**
	✓		76.09	92.47	79.89	84.79	74.93	90.65	72.59	30.08
		✓	75.95	91.79	81.61	85.49	74.97	89.59	71.88	49.78
✓	✓		76.84	92.45	81.19	85.74	75.75	91.76	74.71	26.43
	✓	✓	76.87	93.52	80.88	85.92	75.22	91.31	73.47	29.06
✓		✓	76.49	91.97	81.51	85.94	75.73	90.51	73.85	24.42
✓	✓	✓	**78.04**	**93.93**	**82.55**	**87.11**	**76.92**	**92.64**	**76.37**	25.04

optimization algorithms in key metrics such as mIoU, effectively addressing the challenge of parameter selection caused by variations in individual tree scale.

D. Ablation Experiments

To evaluate the stability of the RtSegNet model and assess the individual contribution and importance of each module, a series of systematic ablation experiments was conducted on the full dataset. Specifically, single-module verification was performed by individually incorporating the GR-DSA, TSILS, and FSGC into the baseline model. Subsequently, pairwise combinations of GR-DSA and TSILS, GR-DSA and FSGC, and TSILS and FSGC were evaluated. Finally, the full RtSegNet architecture integrating all three modules was tested for comparison. To minimize the influence of data variability, all models were trained and evaluated using a fivefold cross-validation strategy. The detailed experimental results are presented in Table IX.

Experimental results show that the GR-DSA, TSILS, and FSGC modules in the proposed method have a positive effect on the recognition ability of the model. As shown in Table IX, the baseline network performed well, but metrics such as accuracy, $F1$-score, and mIoU were lower than those for adding the proposed module. This indicates that there is a limitation in the performance of the baseline network in the rubber tree point cloud segmentation task. The addition of the GR-DSA module improved all evaluation metrics (Cov: +2.84%, mPre: +3.87%, mRec: +5.40%, F-score: +5.08%, Iacc: +3.18%, and mIoU: +5.39%). Obviously, the addition of GR-DSA can help improve the network performance, and the algorithm can better adapt to different sizes of rubber trees by optimizing the clustering parameters, avoiding the oversegmentation and undersegmentation problems caused by fixed parameters. The addition of TSIL to the baseline network also significantly improved the metrics (Cov: +2.74%, mPre: +3.10%, mRec: +4.63%, F-score: +4.86%, Iacc: +3.59%, and mIoU: +5.93%), while the training time was significantly reduced (training efficiency: 62.8%). This shows that the hierarchical incremental processing mechanism of TSIL eliminates redundant calculations and pays more attention to key feature areas. When using FSGC to model spatial relationships, the metrics were also significantly improved (Cov: +2.98%, mPre: +4.55%, F-score: +4.38%, Iacc: +4.24%, and mIoU: +6.10%). At the same time, the best accuracy, $F1$-score, and mIoU were obtained (Cov: +4.93%, mPre: +6.01%, mRec: +6.34%, F-score: +6.70%, Iacc: +6.23%, and mIoU: +9.88%), and the training efficiency was also improved (training time: 46.5%). These results further verify that the module proposed in this article plays an active role in dealing with the task of rubber tree point cloud segmentation.

Fig. 8 shows some visualizations of the ablation experiments on the dataset, where Fig. 8(a) shows the true labeling results. Fig. 8(b) shows the results of baseline network identification, and segmentation breaks and boundary blurring problems can be observed. Compared to Fig. 8(b), the addition of TSIL in Fig. 8(c) results in an improvement in the coherence of segmentation, in addition to a more complete coverage of the target region. As shown in Fig. 8(d), the addition of the TSILS and FSGC combination further improves the boundary accuracy, eliminates more segmentation errors, and makes the boundary positioning more accurate, compared to Fig. 8(b) of the baseline network. Comparing Fig. 8(b) with Fig. 8(e), it is clear that after the full approach, the model focuses more on fine features. The local features of the target are more prominent, and the boundary contour is clearer and more coherent than the background, which is due to the optimization of the clustering parameters by the GR-DSA algorithm, so that rubber trees of different sizes can be properly segmented. Comparative analysis of elliptical marker regions demonstrates the gradual improvement from baseline to the full method, especially in the treatment of complex boundaries and small structures. Finally, as shown in Fig. 8(e), the complete method network improves the accuracy and accuracy of recognition, and achieves a segmentation effect that is highly consistent with the real annotation. Therefore, the complete method network that we propose is very effective.

E. Comparison With SOTA Methods

To validate the performance of the proposed method, we compare it with seven state-of-the-art methods on a self-constructed rubber tree dataset, including the traditional algorithms GMM [47], FCM [48], WSA [49], Region Growing [40], K-means [50], and TMGH [51], as well as the deep learning methods PointGroup [52], Treeiso [53], PointMLP [54], and ForAInet [55]. The experiments are conducted using a fivefold cross validation, and the quantitative results are shown in Table X, and the visualization results are shown in Fig. 9.

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Fig. 8. Visualization results of the ablation experiments. (a) GT of rubber tree point cloud data, (b) segmentation results of the baseline network, (c) results after incorporating the TSILS module, (d) results after incorporating the TSILS+FSGC modules, and (e) segmentation results of the complete model. Different colors represent different segmentation categories, and elliptical markers highlight key improvement areas.

As shown in Table X, the proposed method achieves superior performance across all evaluation metrics, with mIoU reaching 76.37% and mPre reaching 93.93%. Traditional clustering methods perform poorly. For example, K-means achieves only 42.2% mIoU, mainly due to its inability to handle the complex geometric structures of rubber trees. Among other traditional methods, GMM achieves 28.95% mIoU due to poor Gaussian assumptions for irregular tree geometries, FCM reaches 47.11% mIoU through fuzzy clustering but struggles with boundary determination, and TMGH performs best at 69.43% mIoU by incorporating hierarchical structures. Among deep learning methods, PointGroup and PointMLP exhibit significant limitations in overlapping regions. Although ForAInet shows relatively high precision, its recall remains insufficient in complex scenarios. Beyond segmentation accuracy, computational efficiency is crucial for practical deployment. Table XI presents comprehensive efficiency benchmarking against state-of-the-art methods under identical hardware conditions. RtSegNet achieves the fastest inference speed at 24.77 ms per batch, outperforming all competing methods. PointMLP requires 36.44 ms, PointGroup takes 31.63 ms, and ForAInet and TreeIso need 165.56 and 285.39 ms, respectively.

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Fig. 9. Visual comparison of rubber tree point cloud segmentation results across different methods. Each row represents a different test case, with methods arranged from left to right: GT, GMM, FCM, WSA, region growing, K-means, PointGroup, Treeiso, PointMLP, TMGH, ForAINet, and RtSegNet (Ours). Different colors represent distinct tree instances, with circled regions highlighting key segmentation differences.

TABLE X
CONTRAST EXPERIMENTAL RESULTS

Method	Cov(%)	mPre(%)	mRec(%)	F-score(%)	mAP(%)	Iacc(%)	mIoU(%)
GMM	35.38	39.83	48.97	47.57	20.74	32.43	28.95
FCM	40.79	62.44	45.01	52.31	37.65	45.45	47.11
WSA	44.94	41.74	47.27	46.99	48.6	43.87	44.31
Region Growing	56.47	68.52	57.33	66.95	53.41	61.45	48.64
K-means	42.36	53.02	29.53	46.62	58.62	48.78	42.2
PointGroup	63.24	68.51	66.89	67.63	60.93	81.44	60.72
Treeiso	68.66	81.9	75.22	73.48	66.66	77.22	68.83
PointMLP	67.12	81.07	68.11	76.21	67.69	74.52	65.85
TMGH	70.51	74.54	73.83	76.35	67.68	79.09	69.43
ForAInet	73.22	89.79	78.86	81.09	71.4	85.18	73.68
RtSegNet(Ours)	**78.04**	**93.93**	**82.55**	**87.11**	**76.92**	**92.64**	**76.37**

TABLE XI
COMPUTATIONAL EFFICIENCY COMPARISON WITH EXISTING METHODS

Method	Inference Time(ms/batch)	FLOPs(GFLOPs)	MACs(GMACs)	GPU Memory(MB)	CPU Memory(MB)	Parameters(M)
GMM	29.11	2.919	1.459	136.36	207.3	-
PCM	378.67	22.977	7.954	**106.4**	347.5	-
WSA	32.31	16.77	9.38	166.4	156.2	-
RGA	55.39	10.25	5.15	234.33	137.57	-
K-means	79.87	5.79	2.89	273.4	**134.3**	-
PointGroup	31.63	71.33	135.18	331.44	144.54	**11.12**
Treeiso	285.39	**2.84**	**1.47**	166.4	400.5	-
PointMLP	36.44	12.58	6.29	113.88	276.78	12.6
TMGH	192.42	4.59	2.43	265.73	418.42	-
ForAInet	165.56	47.49	23.74	471.53	611.25	11.87
RtSegNet (Ours)	**24.77**	59.04	44.52	507.8	621.57	12.44

This efficiency stems from GR-DSA's optimized clustering parameters and TSILS's incremental learning strategy that eliminates redundant computations. In terms of computational complexity, RtSegNet requires 59.04 GFLOPs, falling in the mid-range among deep learning approaches while demonstrating better efficiency than PointGroup at 71.33

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TABLE XII
SUMMARY OF THE MAIN CHARACTERISTICS OF THE TWO FOREST GENERALIZATION DATASETS USED IN THIS STUDY

Dataset	Country	Forest type and main tree species	n trees and tree density	Mean tree height (m)	Data Type (sensor)	Average point density (pts m^{-2})
FOR-instance test data	Norway, Czech Republic, Austria, New Zealand, Australia	Boreal coniferous forest (*Picea abies*), temperate coniferous forests (*Pinus sylvestris*), temperate mixed deciduous forest (*Fagus sylvatica*, *Acer platanoides*), plantation forest (*Pinus radiata*), dry sclerophyll forest (*Eucalyptus* sp.)	334; 425 trees ha^{-1}	21.6	ULS (Riegl VUX1 UAV and miniVUX)	6747
NIBIO MLS test data	Norway	Boreal coniferous forest (*Picea abies*, *Pinus sylvestris*, *Betula pendula*)	258; 1407 trees ha^{-1}	14.7	MLS (GeoSLAM ZEB-HORIZON)	20,000

GFLOPs. Memory consumption reaches 507.8-MB GPU and 621.57-MB CPU, reflecting the computational demands of graph neural networks and exemplar storage. The model contains 12.44 M parameters, comparable to existing deep learning methods.

Visualization results of each method are illustrated in Fig. 9. One limitation of traditional methods lies in their inability to distinguish individual trees in interlaced canopy regions, where neighboring rubber trees share similar point cloud patterns and geometric features. As a result, these methods tend to merge multiple distinct trees into a single instance. For instance, in the circled regions of Test #1 and Test #2, K-means incorrectly segments two adjacent trees as one instance. GMM exhibits similar undersegmentation issues in overlapping regions due to its Gaussian assumptions being poorly suited for irregular tree geometries. FCM produces relatively smoother clustering results but creates inaccurate boundaries where overlapping crowns have similar fuzzy membership values. TMGH demonstrates better structural preservation through hierarchical relationships but still merges adjacent trees when graph connectivity becomes ambiguous in dense regions. Region Growing demonstrates a similar undersegmentation issue, as these algorithms rely solely on point-wise distance thresholds and fail to differentiate spatially adjacent but structurally distinct instances. In contrast, RtSegNet models deeper spatial topologies among trees. Through graph-based representations, it captures structural differences even in overlapping areas, effectively preventing erroneous merges. In Tests #3 and #4, PointGroup and PointMLP frequently oversegment a single tree into multiple fragments, with the same tree appearing in red, green, and blue. This oversegmentation arises from the limited generalization ability of fixed feature extractors to accommodate diverse tree morphologies. RtSegNet addresses this problem by employing incremental learning, which enables the network to accumulate and refine feature representations of various tree forms over time. As a result, the structural integrity of each instance is well preserved. In scenarios with pronounced scale differences, such as in Tests #5–#7, methods with fixed-parameter settings often miss small-scale trees entirely or excessively subdivide larger ones. RtSegNet introduces dynamic parameter optimization that adjusts clustering sensitivity based on local point density, ensuring balanced segmentation across different tree sizes. While ForAInet performs well in clearly defined regions, RtSegNet exhibits greater robustness in complex overlapping environments. As shown in Tests #6 and #7, it maintains higher recall while preserving boundary precision.

F. Generalization Experiment

To evaluate the generalization ability of RtSegNet in diverse forest environments, two challenging public datasets were selected for testing. A summary of the main forest characteristics in both datasets is provided in Table XII. The FOR-instance dataset [56] covers five national research sites with distinct latitudes, climatic conditions, and forest management practices, while the NIBIO MLS dataset [15] represents boreal coniferous forests. These datasets exhibit substantial differences from the training data in terms of tree species composition, forest structure, and environmental complexity, thereby providing rigorous conditions to assess the model's generalization performance. Both datasets were applied directly to RtSegNet without preprocessing modifications. This preserves inherent dataset differences, including point density (6747 versus 20 000 versus 10 667 pts/m^2), sensor platforms (ULS versus MLS versus UAV), and forest types, which reflect realistic deployment scenarios. Performance variations under these conditions represent genuine algorithmic robustness rather than preprocessing artifacts, providing authentic benchmarks for practical applications where methods must transfer across diverse environments without extensive parameter tuning.

1) Generalization Experiment of for-Instance Dataset: Table XIII shows a comparison of the performance of the different methods at the five test sites. RtSegNet achieved the best performance across all sites, with an F-score of 91.12% for the CULS site. Fig. 10 shows the visual segmentation results for each method. Traditional methods exhibit distinct limitations when applied to diverse forest environments. K-means produces confusing segmentation results with poorly defined tree boundaries across all test sites. GMM demonstrates unstable clustering behavior, particularly struggling at TUWIEN and NIBIO sites where its Gaussian assumptions fail to capture complex tree morphologies. FCM generates smoother clustering outputs but creates inaccurate boundaries

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TABLE XIII
COMPARISON OF GENERALIZATION PERFORMANCE OF DIFFERENT SEGMENTATION METHODS ON THE FOR-INSTANCE DATASET

Method	Test dataset	Cov(%)	mPre(%)	mRec(%)	F-score(%)	mAP(%)	Iacc(%)	mIoU(%)
GMM	FOR-instance CULS	27.42	24.93	46.43	30.48	26.32	25.67	14.87
	FOR-instance NIBIO	21.73	14.52	34.94	24.75	14.37	18.53	13.01
	FOR-instance TUWIEN	19.81	26.91	36.73	39.43	11.25	22.86	16.21
	FOR-instance SCION	21.09	20.78	47.83	28.01	18.54	23.36	20.92
	FOR-instance RMIT	20.5	47.18	25.83	25.19	16.09	33.12	26.96
FCM	FOR-instance CULS	28.94	38.41	51.26	47.21	27.31	42.48	26.34
	FOR-instance NIBIO	14.35	17.35	50.71	27.86	16.86	21.25	14.87
	FOR-instance TUWIEN	22.18	38.02	59.08	49.05	17.98	35.83	17.45
	FOR-instance SCION	29.26	23.98	49.78	32.37	26.92	26.92	24.06
	FOR-instance RMIT	23.08	53.95	38.10	35.42	28.99	43.75	36.77
WSA	FOR-instance CULS	45.27	37.31	51.55	35.91	27.5	47.54	26.56
	FOR-instance NIBIO	25.15	27.17	39.87	28.98	18.64	33.54	15.94
	FOR-instance TUWIEN	36.25	29.7	45.38	33.82	20.05	49.69	28.73
	FOR-instance SCION	44.48	31.64	58.54	41.08	40.23	62.59	39.33
	FOR-instance RMIT	35.62	25.12	54.9	32.85	32.91	49.67	31.46
Region Growing	FOR-instance CULS	69.48	84.41	69.98	74.73	57.76	74.7	55.78
	FOR-instance NIBIO	42.12	66.92	44.72	49.14	24.91	43.07	49.14
	FOR-instance TUWIEN	49.40	61.06	51.5	49.58	16.99	59.31	56.41
	FOR-instance SCION	28.63	23.52	24.18	21.24	17.69	29.41	23.79
	FOR-instance RMIT	34.53	61.46	33.26	39.99	25.64	56.21	39.9
K-means	FOR-instance CULS	30.82	48.05	28.78	36.04	54.54	31.3	46.11
	FOR-instance NIBIO	38.75	29.75	43.22	25.48	38.92	35.18	34.67
	FOR-instance TUWIEN	25.44	39.52	24.11	26.05	21.71	26.78	17.93
	FOR-instance SCION	24.51	14.85	38.15	21.38	24.11	25.15	24.33
	FOR-instance RMIT	37.5	44.03	32.8	37.59	48.5	37.85	38.5
PointGroup	FOR-instance CULS	71.1	80.28	74.53	74.63	52.32	80.28	66.34
	FOR-instance NIBIO	42.27	61.47	43.84	45.25	23.09	61.47	45.22
	FOR-instance TUWIEN	49.01	59.87	51.4	48.49	25.6	58.53	43.35
	FOR-instance SCION	40.21	40.09	39.24	34.87	11.64	41.59	36.42
	FOR-instance RMIT	35.7	56.47	33.42	36.9	13.14	54.4	34.64
Treeiso	FOR-instance CULS	79.36	51.48	86.81	76.2	56.53	79.04	60.21
	FOR-instance NIBIO	58.57	52.08	62.65	56.88	58.53	69.93	47.3
	FOR-instance TUWIEN	48.12	45.94	53.53	39.01	37.39	45.69	35.99
	FOR-instance SCION	72.42	68.1	70.98	56.59	43.18	72.18	45.76
	FOR-instance RMIT	50.31	64.46	48.66	41.17	47.65	50.90	32.5
PointMLP	FOR-instance CULS	59.99	81.29	61.17	67.06	44.31	75.17	53.58
	FOR-instance NIBIO	47.53	59.21	51.32	51.59	21.06	59.21	41.42
	FOR-instance TUWIEN	52.94	54.25	55.85	50.23	17.79	53.07	46.85
	FOR-instance SCION	33.20	39.19	28.98	32.28	20.58	39.19	33.85
	FOR-instance RMIT	28.11	54.68	26.34	32.51	17.1	49.02	32.65
TMGH	FOR-instance CULS	75.62	82.47	79.29	80.95	52.38	84.13	68.83
	FOR-instance NIBIO	60.44	71.83	59.51	69.77	48.36	78.35	63.99
	FOR-instance TUWIEN	57.15	62.53	53.89	59.30	38.09	71.44	57.87
	FOR-instance SCION	43.75	70.48	55.23	61.93	35.17	73.46	69.88
	FOR-instance RMIT	56.55	53.51	32.55	62.50	44.81	49.81	53.43
ForAInet	FOR-instance CULS	80.76	**93.98**	84.6	86.96	67.82	90.54	70.78
	FOR-instance NIBIO	62.48	89.4	**67.87**	74.59	54.17	87.27	71.59
	FOR-instance TUWIEN	63.54	71.32	**62.84**	65.49	40.4	75.35	**69.28**
	FOR-instance SCION	69.41	61.23	88.37	72.38	49.53	**77.84**	**72.6**
	FOR-instance RMIT	**63.66**	76.55	57.71	62.56	48.44	84.8	62.92
RtSegNet (Ours)	FOR-instance CULS	**91.89**	91.8	**94.2**	**91.12**	**69.87**	**91.8**	**75.61**
	FOR-instance NIBIO	**65.95**	85.71	**69.87**	**76.96**	**55.79**	**90.71**	**74.55**
	FOR-instance TUWIEN	**67.2**	67.38	**65.13**	**66.81**	**43.53**	**76.97**	67.9
	FOR-instance SCION	**77.96**	**76.28**	83.31	**77.21**	**51.6**	76.28	70.1
	FOR-instance RMIT	62.39	77.13	**58.75**	**63.39**	**48.58**	**76.51**	**64.73**

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TABLE XIV
COMPARISON OF GENERALIZATION PERFORMANCE OF DIFFERENT SEGMENTATION METHODS ON THE NIBIO MLS DATASET

Method	Cov(%)	mPre(%)	mRec(%)	F-score(%)	mAP(%)	Iacc(%)	mIoU(%)
GMM	16.67	23.13	24.48	26.81	16.27	15.97	19.48
FCM	25.44	32.14	32.98	34.64	35.29	28.65	22.58
WSA	23.33	29.79	12.97	26.16	23.33	21.89	19.57
Region Growing	34.75	48.33	32.83	35.95	30.71	48.33	35.94
K-means	17.14	12.75	15.11	20.58	19.45	17.45	18.09
PointGroup	54.75	68.33	52.83	55.95	34.93	68.33	55.94
Treeiso	33.33	56.34	32.1	40.9	51.43	50.92	54.28
PointMLP	34.4	59.07	31.45	47.22	30.69	55.66	41.54
TMGH	42.37	68.94	44.86	60.13	46.17	57.79	54.32
ForAInet	64.22	76.64	58.88	63.68	48.24	74.6	61.83
RtSegNet(Ours)	**73.87**	**81.58**	**61.78**	**68.28**	**61.84**	**81.58**	**65.14**

at CULS and RMIT sites. TMGH demonstrates relatively better boundary preservation among traditional methods but still struggles with tree separation at NIBIO and TUWIEN sites. As can be seen from the annotated areas, deep learning methods also face site-specific challenges. PointGroup frequently incorrectly merges adjacent trees at CULS and NIBIO sites, which is manifested by multiple independent trees being marked with the same color. PointMLP has an oversegmentation problem at the TUWIEN site, where single trees are split into multiple fragments of different colors. RtSegNet maintains segmentation integrity at each site. However, there is a small amount of boundary blurring in the segmentation results at the RMIT site, and the segmentation accuracy of some canopy edge areas needs to be improved. In addition, the recall rate of RtSegNet is relatively low at the SCION site with low point cloud density, indicating that there is still room for improvement in the sparse point cloud environment.

2) Generalization Experiment of NIBIO MLS Dataset: The NIBIO MLS dataset tests the ability to generalize across ecosystems from tropical rubber trees to boreal coniferous forests. Table XIV shows that the performance of all methods on this dataset is degraded to varying degrees. The F-score of RtSegNet is 68.28%, which is 4.6% higher than that of ForAInet. The visualization in Fig. 11 shows that the pinnacle-like structure of conifers poses a challenge to all approaches. GMM has difficulty modeling the vertically elongated crown structures of conifers, as its spherical clustering assumptions are poorly suited for the cone-shaped tree architecture. FCM struggles with the low point density in coniferous canopies, where the sparse needle distribution makes it difficult to determine which points belong to which tree, resulting in poor boundary definition. TMGH demonstrates better structural preservation but struggles with hierarchical clustering in sparse coniferous canopies, where the limited point connectivity disrupts the graph structure needed for effective tree separation. PointGroup and PointMLP showed significant segmentation breaks when dealing with the slender branches of conifers, and the trunk and crown parts were often separated. Although the overall performance of RtSegNet was better, the segmentation accuracy was still not satisfactory in the top areas of some tall conifers, and some of the tips were missed. It is important to note that while the performance of all methods is degraded on the taiga dataset, the performance degradation of RtSegNet is relatively small. These results indicate that RtSegNet has better cross-species adaptability, but there is still room for improvement in the segmentation accuracy in complex environments.

G. Structural Parameter Extraction

In this study, four rubber forests were selected in three different regional environments in Danzhou City, Hainan Province, and each forest area contained about 200 rubber trees. Based on the results of network segmentation, the structural parameters of rubber trees were measured and analyzed, focusing on three key indicators: tree height, canopy diameter, and canopy volume. Fig. 12 shows a scatterplot analysis of the predicted and reference values. The experimental results show that the tree height estimation maintains a very high correlation among the four regions, with R^2 reaching 0.94 and RMSE reaching only 0.52, which proves the stability and accuracy of the model in tree height prediction. The correlation between canopy diameter and canopy volume reached R^2 of 0.79 and 0.80, respectively, which also showed good predictive ability.

The scatter plots of the parameter characteristics of each region are shown in Fig. 13. The results of regional analysis show a strong linear correlation between the predicted value and the actual tree height, and the R^2 is close to 1, which further verifies the stability of the model to estimate the tree height. In terms of crown amplitude measurement, the data points in region #2 were relatively scattered, and the degree of deviation from the ideal fitting line was more prominent than that in other regions, which was speculated to be related to the large individual differences of rubber trees in this area and the complexity of growth conditions. Although the correlation of each region is within the acceptable range, the prediction accuracy of this parameter still has potential for improvement due to the integrity of the labeled data. Compared with the existing methods, the segmentation network proposed in this study shows outstanding performance in the field of automatic parameter extraction. The algorithm has achieved

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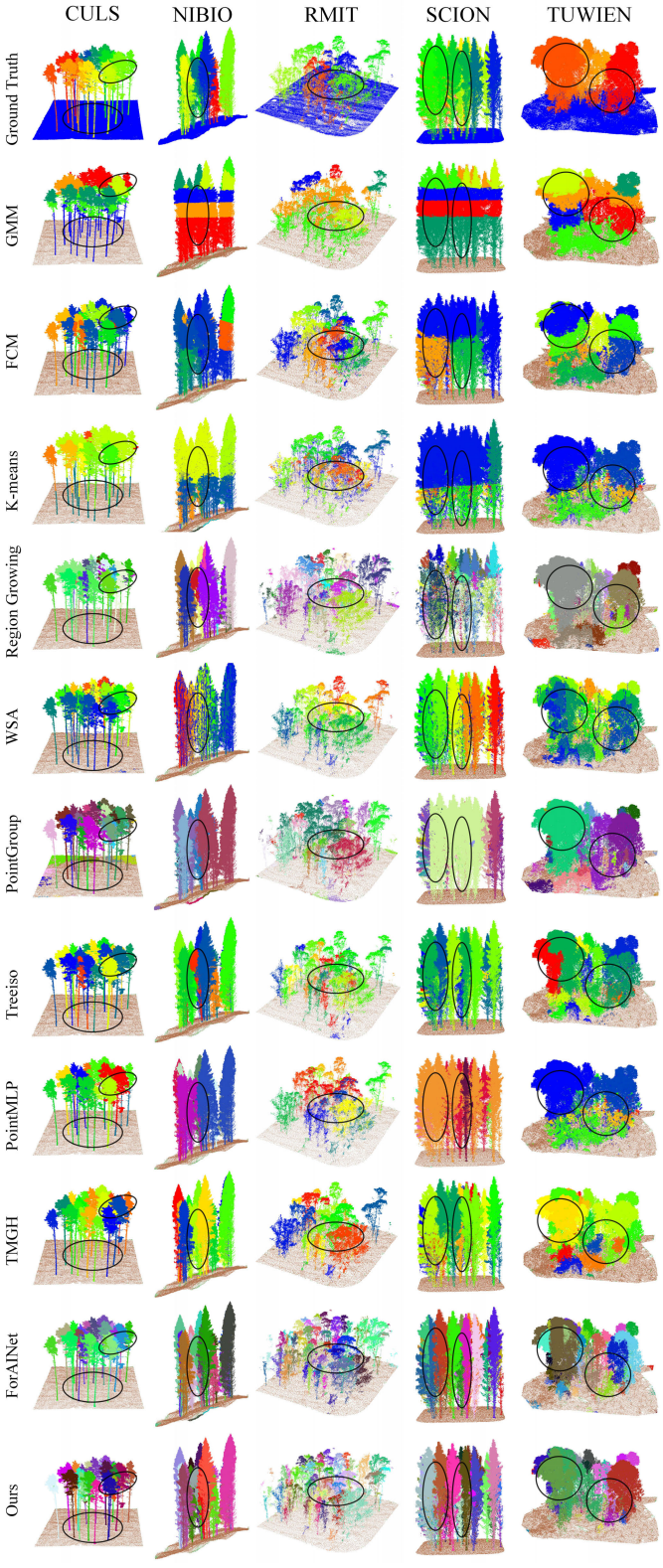


Fig. 10. Visual comparison results of different tree segmentation methods on the FOR-instance dataset. The five columns correspond to test samples from five national sites in the FOR-instance dataset (CULS, NIBIO, RMIT, SCION, and TUWIEN). The nine rows represent the GT annotations, segmentation results of GMM, FCM, K-means, Region Growing, WSA, PointGroup, Treeiso, PointMLP, TMGH, ForAINet, and the proposed method, from top to bottom. The circled regions highlight key differences for comparison.

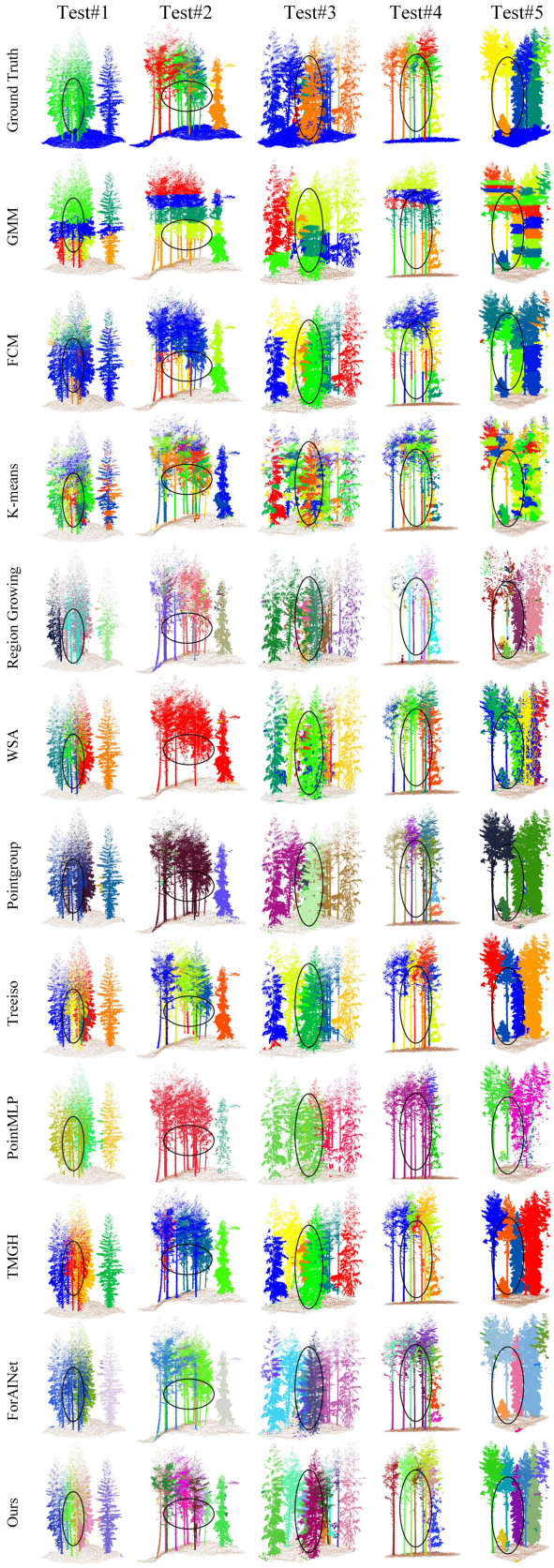


Fig. 11. Visual comparison of different tree segmentation methods on the NIBIO MLS dataset. The five columns correspond to the five test samples in the NIBIO MLS dataset, and the nine rows from top to bottom are the segmentation results of real annotation, GMM, FCM, K-means, Region Growing, WSA, PointGroup, Treeiso, PointMLP, TMGH, ForAINet, and the method in this article.

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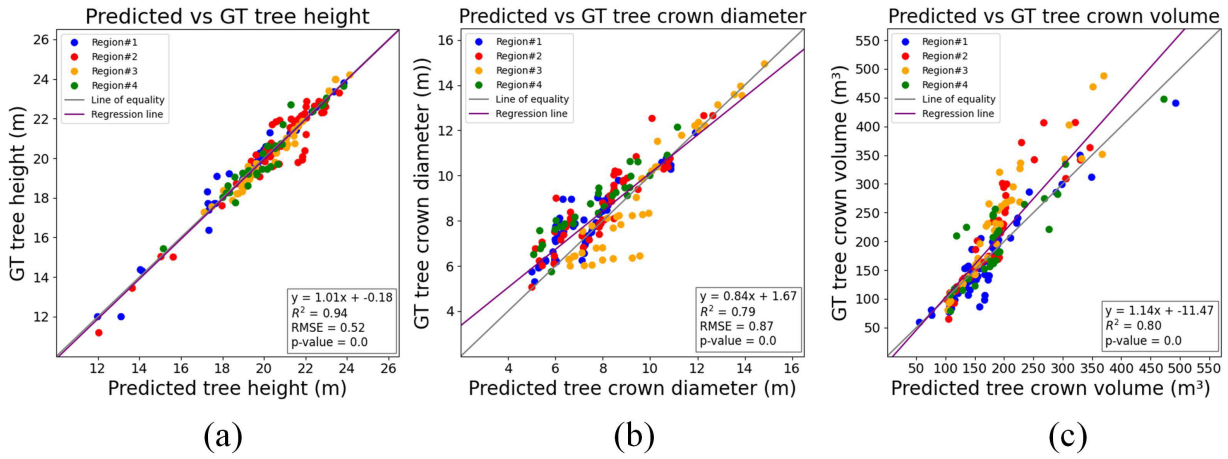


Fig. 12. Scatter plots comparing predicted versus ground truth (GT) measurements for rubber tree structural parameters across four regions in Danzhou City, Hainan Province. (a) Tree height predictions demonstrating high correlation and accuracy with the smallest prediction error among the three parameters. (b) Tree crown diameter predictions showing good predictive performance across all regions. (c) Tree crown volume predictions also exhibiting strong correlation between predicted and measured values. Data points are color-coded by region (Regions 1-4), with the purple line indicating the line of equality and the black dashed line showing the regression line.

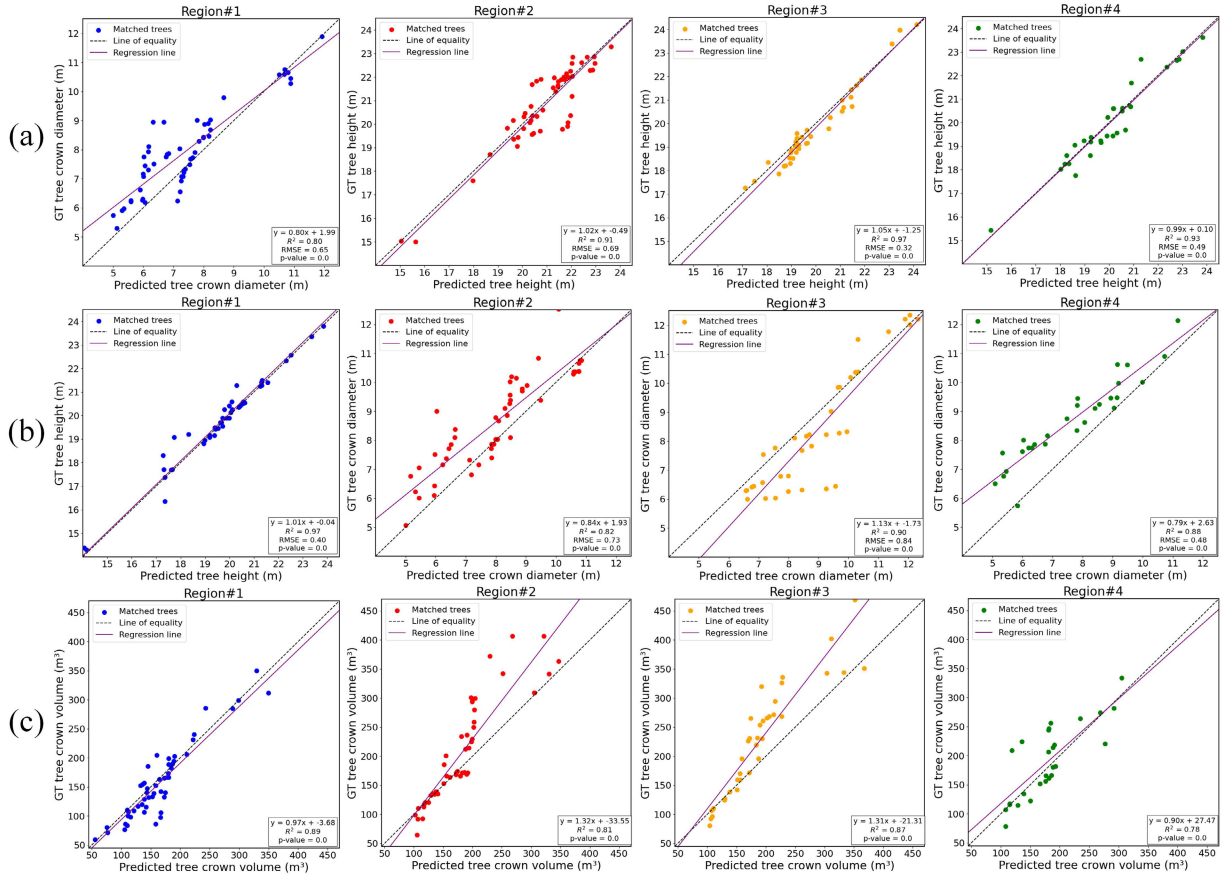


Fig. 13. Regional scatter plot analysis of predicted versus ground truth (GT) measurements for rubber tree structural parameters across four regions in Danzhou City, Hainan Province. (a) Tree height predictions for each individual region. (b) Tree crown diameter predictions for each individual region. (c) Tree crown volume predictions for each individual region.

satisfactory results in the accuracy (R^2 = 0.94) and error level (RMSE = 0.52) of tree height measurement, which lays a technical foundation for the construction of intelligent forestry management system. The cross-environmental stability of this technical scheme provides a new technical path for the monitoring application of large-scale rubber plantations.

H. Practical Application Analysis

The test data in the original dataset were insufficient to comprehensively evaluate the performance of RtSegNet in rubber tree point cloud segmentation. To further assess the model's effectiveness in real-world segmentation tasks, we employed a DJIM300RTK UAV equipped with a CBI-Lite LiDAR sensor to acquire real-time point cloud data of rubber plantations in

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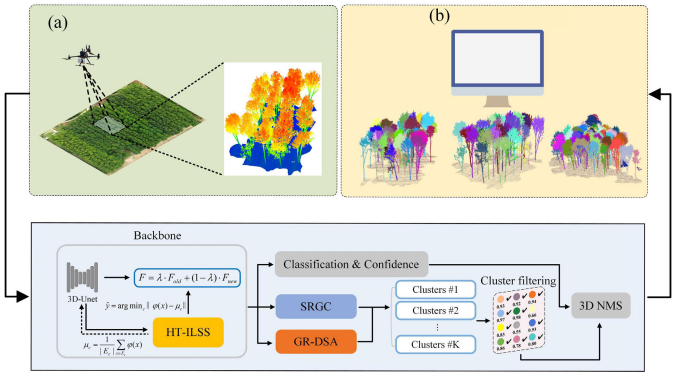


Fig. 14. Workflow for a practical application. (a) Point cloud data acquisition module. (b) Host computer.

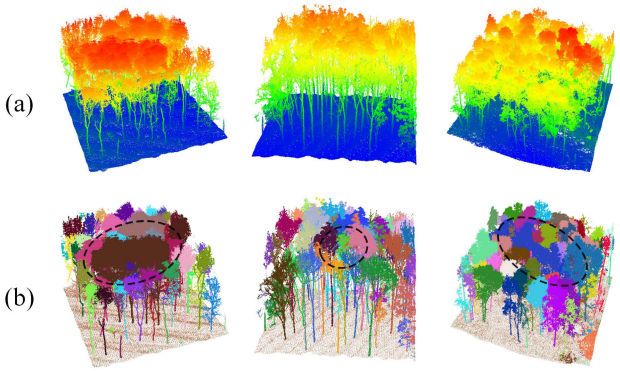


Fig. 15. Failure case in dense overlapping environments. (a) Height-rendered original point cloud and (b) segmentation results, where different colors represent distinct tree instances predicted by the network, and circled regions highlight incorrect merging of separate trees into single instances.

Danzhou, Hainan Province, as shown in Fig. 14(a). In total, point cloud data from 20 plantation blocks were collected. The data were subsequently processed and fed into the RtSegNet model following the workflow illustrated in the architectural diagram below Fig. 14. The segmentation pipeline involved feature extraction using the backbone network, followed by individual tree instance separation based on a classification-confidence module and a clustering-based filtering algorithm. Fig. 14(b) presents the processed results on the host terminal, where colorized point clouds clearly delineate the boundaries of individual rubber trees. The segmentation outcomes were rendered on-screen, enabling precise identification of structural features. Notably, even under challenging conditions such as dense canopy coverage and complex terrain, the proposed method maintained high boundary sensitivity and successfully achieved accurate instance-level segmentation. Overall, the RtSegNet-based rubber tree segmentation algorithm demonstrated strong adaptability and high precision across varying plantation environments.

V. Discussion and Shortcomings

The RtSegNet proposed in this study shows significant advantages in the segmentation of rubber tree point cloud instances. As shown by the experimental results, RtSegNet outperformed the existing methods in all evaluation indicators, with mIoU reaching 76.37% and mPre reaching 93.93%, making breakthroughs in addressing key challenges such as species diversity, crown overlap, and parameter sensitivity of rubber trees.

Despite these achievements, there is still a lot of room for improvement in the RtSegNet model. From the perspective of model complexity, although the three modules of TSILS, FSGC, and GR-DSA improve the segmentation accuracy, they also increase the computational complexity and the difficulty of parameter adjustment. Second, from the perspective of cross-environment adaptability, the generalization experiment results show that the performance of RtSegNet in heterogeneous environments such as taiga decreases significantly, especially on the NIBIO MLS dataset; the F-score decreases from 87.11% to 68.28%, indicating that the model has a strong dependence on the training environment. It is worth noting that the recall rate of RtSegNet is relatively low when dealing with scenarios with low point cloud density (such as SCION sites), and the segmentation accuracy of the top region of conifer spire-like structures is not ideal, which indicates that the algorithm still needs to be further optimized when dealing with sparse point clouds and special geometries. As shown in Fig. 15, in extremely dense environments where multiple trees highly overlap [Fig. 15(a)], although the model can identify the main tree structures, it still faces challenges in boundary regions with severe crown overlap, resulting in incorrect merging of distinct trees into single instances.

We found that the performance of RtSegNet is largely limited by the quality of the training data distribution and labeling, particularly in scenarios with extreme crown overlap density [Fig. 15(b)]. Although the model performs well on the rubber tree dataset, the incorrect merging observed in densely overlapping environments indicates that the generalization ability of the model is still limited when encountering morphological complexities beyond the training distribution, especially in areas where the spatial arrangement and crown interaction patterns significantly differ from the training data. The spectral clustering component in FSGC struggles to maintain discriminative power when trees exhibit nearly identical geometric characteristics within overlapping regions, as the frequency domain transformation cannot effectively distinguish between structurally similar instances. This failure stems from the inherent limitation of graph-based approaches when spatial proximity dominates structural differences, causing the FSGC module to establish strong connections between geometrically similar but distinct instances. In addition, although the GR-DSA algorithm solves the problem of parameter adaptation, its convergence speed and stability in handling such extreme density variations with severely overlapping crowns still need to be further verified, as the optimization landscape becomes more complex in these challenging scenarios.

Future research directions should focus on improving the generalization ability of models across environments and developing more lightweight network architectures to reduce computing costs. At the same time, it is expected to further improve the segmentation accuracy and robustness in complex forest environments by exploring the multimodal data fusion strategy and combining the complementarity of multisource data such as RGB images and elevation information. In addition, the establishment of a more diverse training dataset,

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covering different tree species, growth environments, and point cloud quality conditions, will help improve the universality of the model.

VI. Conclusion

This article proposes a novel point cloud segmentation network, termed RtSegNet, which enables accurate individual tree identification and structural parameter extraction. The main conclusions are as follows.

1) The three core modules incorporated in RtSegNet each provide distinct functionalities. The TSILS module adopts an incremental learning framework with a tree-structured sampling strategy, effectively addressing species diversity in rubber plantations. It enhances generalization performance while reducing training time. The FSGC module leverages graph neural networks to model spatial topological relationships and integrates Fourier spectral fusion to substantially improve segmentation accuracy in overlapping canopy regions. The GR-DSA algorithm introduces a GRW mechanism to optimize clustering parameter selection, thereby overcoming the limitations of traditional fixed-parameter strategies.
2) Comprehensive performance evaluations demonstrate that RtSegNet achieves outstanding results in rubber tree point cloud segmentation, with an mIoU of 76.37% and a mean instance precision (mPre) of 93.93%, outperforming state-of-the-art methods. The tree height estimation achieved an R^2 of 0.94 and an RMSE of only 0.52, offering a reliable solution for the precise measurement of forestry parameters. These results provide a solid technical foundation for resource inventory, growth monitoring, and refined management of rubber plantations.

Nevertheless, there remain areas for improvement in terms of environmental adaptability and computational efficiency. Future research will focus on enhancing the model's adaptability across diverse tree species to improve generalizability. Additionally, efforts will be directed toward exploring cross-domain multimodal data fusion approaches that integrate geometric features from point clouds with complementary information from RGB imagery and elevation data, thereby advancing the practical application of point cloud segmentation techniques in heterogeneous forest environments.

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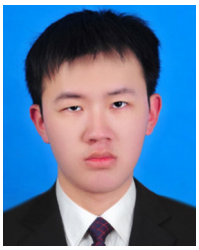
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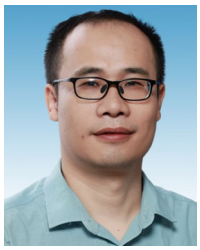
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